NOVINCORP Virtualized Cisco + VMware Lab

English Technical Documentation | Resume / Portfolio Edition

Small-Enterprise Data Center, Cisco Campus Network, Active Directory, Shared iSCSI Storage and High Availability

Document purpose

This document records the design decisions, implementation sequence, network architecture, virtualization layout, Active Directory/file services, security controls, validation tests and troubleshooting lessons of the lab. It is written to be reusable as a technical portfolio / resume case study.

Current lab status: core virtualization, shared iSCSI storage, AD/DNS/DHCP, VLAN routing, ACLs, file permissions and vCenter HA validation are operational. Internet edge firewall/VPN is planned as the next phase.

Item	Current implementation
Host platform	VMware Workstation 17
Hypervisors	2 × ESXi 6.7
Virtual management	VCSA 6.7
Storage	TrueNAS SCALE 25.10.1 + iSCSI shared datastore
Network emulation	EVE-NG 6.2; IOL L2/L3 roles
Server OS	Windows Server 2016 (DC / DNS / DHCP / File Server; WEB role as designed)

Contents

How to use this document

Capture the final command outputs and screenshots listed in Sections 16, 18 and Appendix A after the final hardening pass. Replace any bracketed placeholder values with the final lab value before publishing externally.

1. Executive Summary

This project is a small-enterprise, resume-oriented lab that combines a Cisco Layer-3 campus network with a two-host VMware virtualization environment, shared iSCSI storage, Windows Server infrastructure services and a centralized Active Directory security model. The lab intentionally separates user, server, storage, management and vMotion traffic into dedicated VLANs and uses the Cisco core for inter-VLAN routing and access control.

The virtualization stack uses VMware Workstation 17 as the physical lab platform, two nested ESXi 6.7 hosts and VCSA 6.7 for centralized management. TrueNAS SCALE 25.10.1 provides shared iSCSI storage. Windows Server 2016 provides AD DS, DNS, DHCP and file services. EVE-NG 6.2 hosts the Cisco switching/routing simulation; the final stable access-switch image used in the lab is i86bi_linux_l2-adventerprisek9-ms.SSA.high_iron_20190423.bin.

Resume-ready technical themes

- Designed a segmented small-enterprise network using VLANs 10/20/30/40/50/60/70 with L3 SVIs on the core switch.
- Implemented DHCP relay from client VLANs to a centralized Windows Server 2016 DHCP service.
- Built a two-host nested ESXi 6.7 environment with VCSA 6.7, standard virtual switches, dedicated VMkernel traffic and shared iSCSI storage.
- Integrated TrueNAS iSCSI with ESXi and validated a shared VMFS datastore capable of supporting vCenter-managed workloads and HA tests.
- Implemented role-based Active Directory groups, GPOs and NTFS/SMB permission models including per-department shares and Creator Owner logic for a shared Public folder.
- Designed layered access control so IT administration is privileged while Sales/Staff receive only required access to the DC and their own user segment.
- Performed packet-level and throughput troubleshooting using Wireshark, ICMP, TCP/445 and iperf3; migrated the network emulation layer from GNS3 to EVE-NG after instability was traced to the virtual switching/emulation layer.

Design principles

- Management traffic is separated from user and data traffic.
- Storage/iSCSI traffic is isolated from normal client/server traffic.
- Inter-VLAN routing and policy enforcement stay centralized on the core switch to keep the lab simple and deterministic.
- Virtualization features are introduced only where they add value to the lab objective; unsupported/fragile features such as FT were intentionally omitted.

2. Project Goals and Design Constraints

2.1 Functional goals

- Provide isolated Sales, IT and Staff user networks with controlled inter-VLAN access.
- Provide centralized AD DS, DNS and DHCP services from a Windows Server 2016 DC.
- Provide two ESXi hosts managed by VCSA.
- Provide shared iSCSI storage from TrueNAS so VMs can be restarted on the surviving ESXi host during HA testing.
- Maintain independent Management, Storage and vMotion networks.
- Use only the minimum lab hardware needed to demonstrate operational concepts.

2.2 Resource constraints

Constraint	Budget	Design consequence
Host memory	~45-46 GB	ESXi hosts sized at ~15 GB each; VMs kept lightweight
SSD capacity	~250 GB	Thin/sparse provisioning used where safe; shared iSCSI datastore for VM files
Hypervisor version	ESXi 6.7	Nested lab design and virtual NIC behavior must be kept conservative
Lab-only objective	Functional validation	Performance is measured comparatively; not treated as production-grade capacity

3. Technology Stack and Lab Resource Budget

Component	Version / Type	Qty	Notes
VMware Workstation	17	1	Physical lab platform
ESXi	6.7	2	Nested hosts
VCSA	6.7	1	Central management
TrueNAS	SCALE 25.10.1	1	iSCSI storage VM
Windows Server	2016	1	DC
Cisco L2/L3 emulation	EVE-NG 6.2 / IOL	2 logical roles	Core + Access
User segments	VLAN 10/20/30	3	Sales / IT / Staff
Infra segments	VLAN 40/50/60/70	4	Servers / Storage / Mgmt / vMotion
Workstation RAM	~45 GB total	1	Nested environment

3.1 Current VM resource notes

- ESXi01: approximately 15 GB RAM; local ESXi disk is small and treated as host/boot storage rather than the primary VM datastore.
- ESXi02: approximately 15 GB RAM; same storage strategy.
- TrueNAS: 2 vCPU / 4 GB RAM; approximately 170 GB data disk used for the lab storage pool/zvol.
- VCSA: 10 GB RAM was retained after validation showed the lab VCSA build was not reliable at 8 GB.
- DC and WEB: intentionally lightweight Windows Server 2016 allocations to fit the lab memory budget.

4. Logical Network Architecture

The following diagram represents the intended separation of the user and infrastructure planes. The core switch owns the default gateways and inter-VLAN routing; the access switch is Layer 2 only.

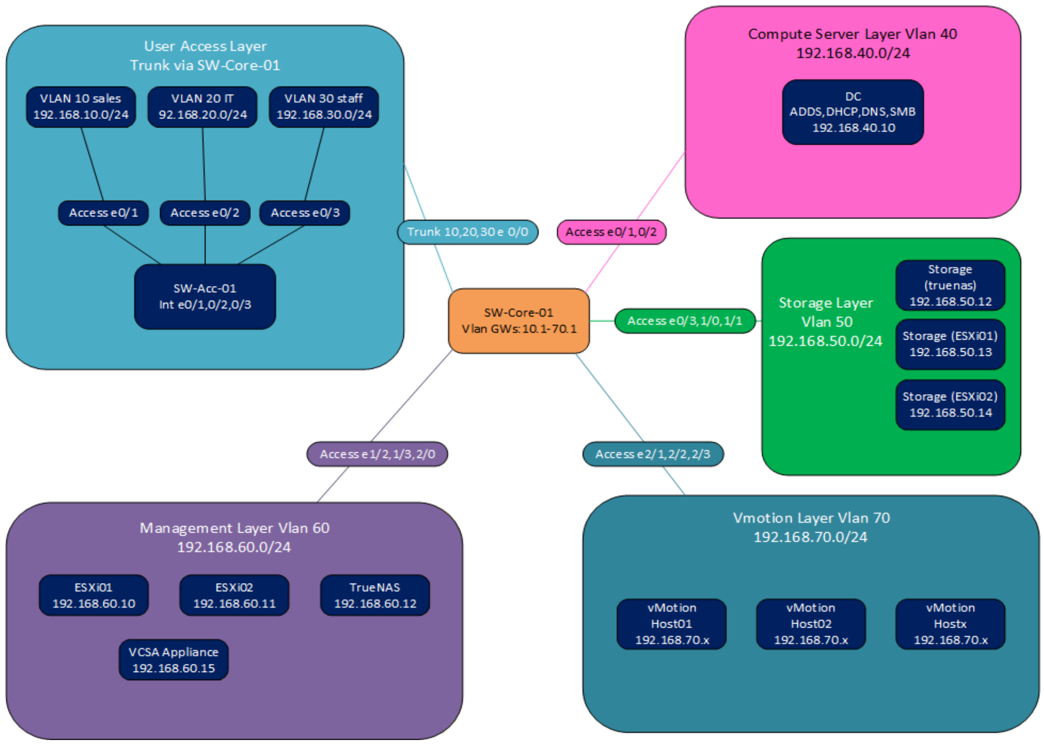


Figure 1 - High-level virtualization and network architecture

4.1 VLAN security model

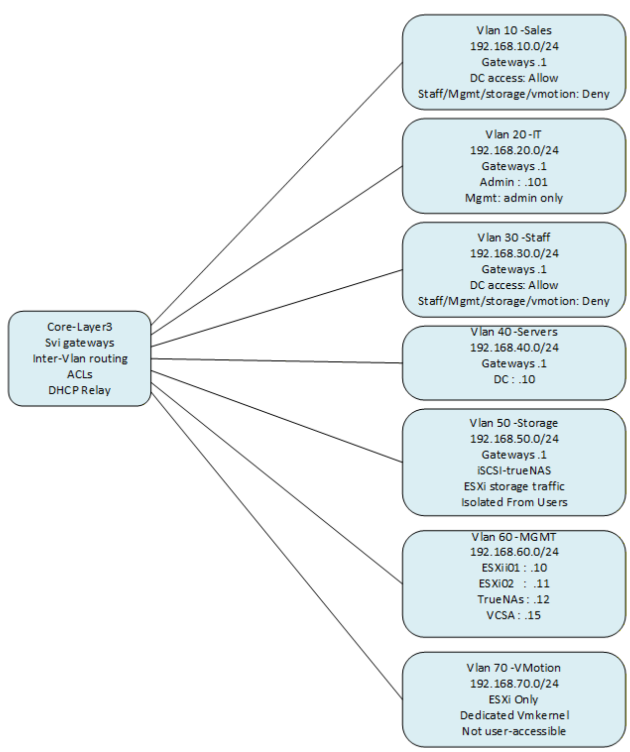


Figure 2 - VLAN roles and security boundaries

4.2 Key design decisions

- VLAN 10/20/30 are user VLANs. Only one trunk connects the access switch to the core for these user VLANs.
- VLAN 40 is the Server VLAN and hosts the DC and WEB workloads.
- VLAN 50 is the iSCSI Storage VLAN.
- VLAN 60 is the Management VLAN for ESXi, VCSA and TrueNAS management access.
- VLAN 70 is reserved for vMotion.
- ESXi Management is separated from the data trunk so loss of a data path does not automatically remove management access.

5. VLAN and IP Addressing Plan

VLAN	Name	Subnet	Gateway	Primary use	Notes
10	SALES	192.168.10.0/24	192.168.10.1	Sales clients	Restricted east-west
20	IT	192.168.20.0/24	192.168.20.1	IT clients/admin	Admin IP 192.168.20.101
30	STAFF	192.168.30.0/24	192.168.30.1	Staff clients	Restricted east-west
40	SERVERS	192.168.40.0/24	192.168.40.1	DC / WEB	DC 192.168.40.10
50	STORAGE	192.168.50.0/24	192.168.50.1	iSCSI	ESXi storage traffic
60	MANAGEMENT	192.168.60.0/24	192.168.60.1	ESXi/VCSA/TrueNAS Mgmt	Admin-only from IT
70	VMOTION	192.168.70.0/24	192.168.70.1	vMotion	ESXi-only

5.1 Fixed infrastructure addresses

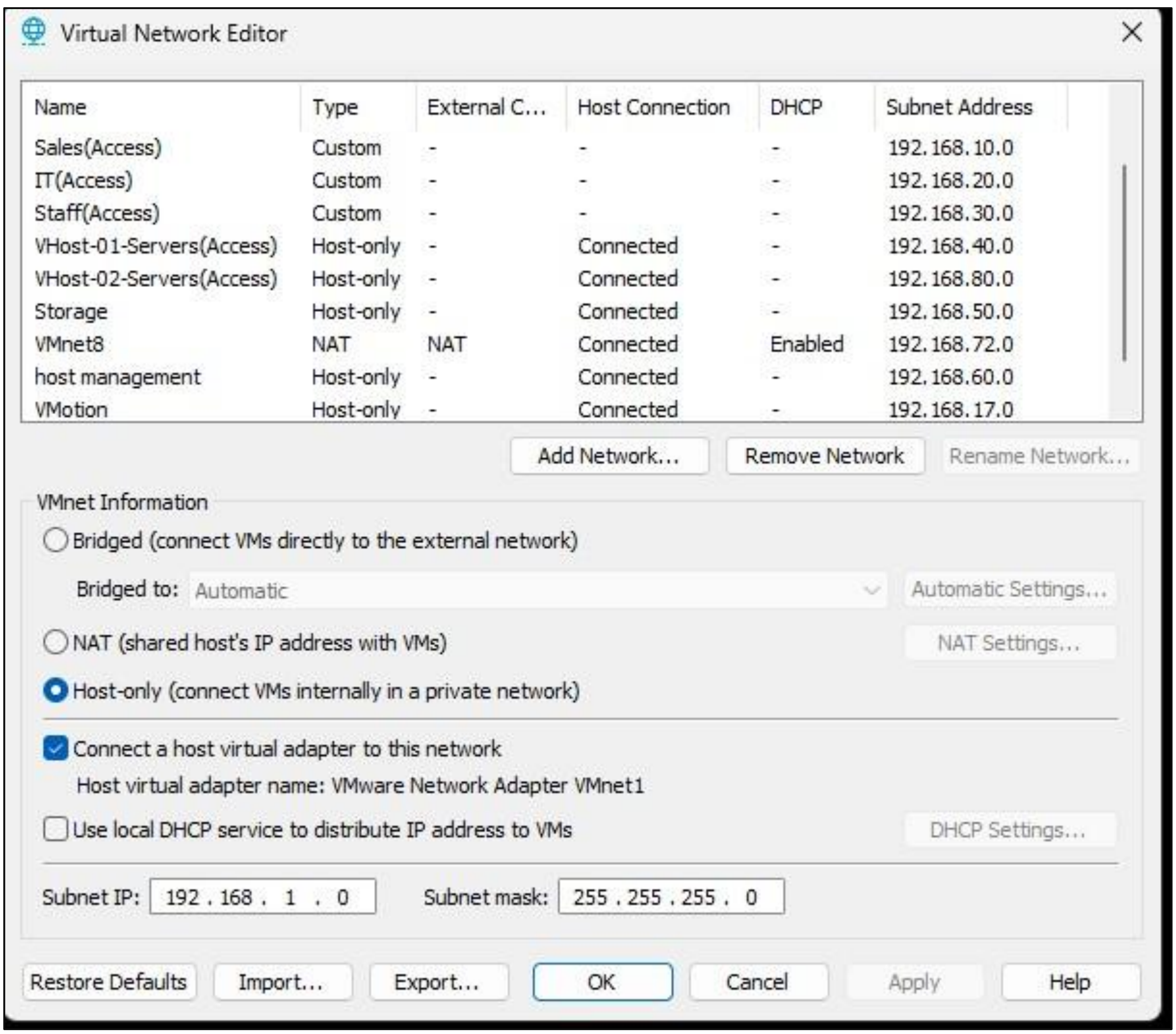
Device	IP address	Role
ESXi01	192.168.60.10	Management
ESXi02	192.168.60.11	Management
TrueNAS	192.168.60.12	Management
VCSA	192.168.60.15	Management
DC	192.168.40.10	AD DS / DNS / DHCP / File Server
IT Admin workstation	192.168.20.101	Privileged management source
Host Admin	192.168.60.100	Managed all vms from host
SW-Acc-01	192.168.20.10	Access switch management

6. VMware Workstation and EVE-NG Network Design

The lab initially used GNS3 for the Cisco layer. During validation, IOSv/E1000 instability caused CPU-hog, CPU exception and packet-loss behavior. The Cisco emulation layer was migrated to EVE-NG 6.2. The migration retained the same logical VLAN/IP architecture while improving practical network stability.

6.1 Workstation virtual networking

Network	Purpose	Typical connection	DHCP
VMnet-MGMT	ESXi/infra management	Host + ESXi+truenas mgmt NICs	Off
VMnet-SALES	Sales endpoint bridge	Sales client ↔ EVE Cloud/Pnet	Off
VMnet-IT	IT endpoint bridge	IT client ↔ EVE Cloud/Pnet	Off
VMnet-STAFF	Staff endpoint bridge	Staff client ↔ EVE Cloud/Pnet	Off
VMnet-Host01-SERVERS	ESXi01 server data	EVE ↔ ESXi01 vmnic1	Off
VMnet-Host02-SERVERS	ESXi02 server data	EVE ↔ ESXi02 vmnic1	Off
VMnet-STORAGE	Direct iSCSI path	TrueNAS storage NIC ↔ ESXi storage VMkernel	Off



EVE-NG VMNETs

Network Adapter	Custom (VMnet0)
Network Adapter 2	Custom (Sales(Access))
Network Adapter 3	Custom (IT(Access))
Network Adapter 4	Custom (Staff(Access))
Network Adapter 5	Custom (VHost-01-Servers(Ac...
Network Adapter 6	Custom (VHost-02-Servers(Ac...
Network Adapter 7	Custom (Storage)
Network Adapter 8	Custom (host management)
Network Adapter 9	Custom (VMotion)

TrueNAS VMNETs

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (IDE)	170 GB
Hard Disk 2 (IDE)	8 GB
CD/DVD (IDE)	Using file E:\Softwares\files\I...
Network Adapter	Custom (host management)
Network Adapter 2	Custom (Storage)
Sound Card	Auto detect
Display	Auto detect

ESXi01 VMNETs

Memory	15 GB
Processors	4
Hard Disk (SCSI)	10 GB (Preallocated)
CD/DVD (IDE)	Using file E:\Softwares\files\I...
Network Adapter	Custom (host management)
Network Adapter 2	Custom (VHost-01-Servers(Ac...
Network Adapter 3	Custom (Storage)
Network Adapter 4	Custom (VMotion)
USB Controller	Present
Display	Auto detect

ESXi02 VMNETs

Memory	15 GB
Processors	4
Hard Disk (SCSI)	10 GB (Preallocated)
CD/DVD (IDE)	Using file E:\Softwares\files\I...
Network Adapter	Custom (host management)
Network Adapter 2	Custom (VHost-02-Servers(Ac...
Network Adapter 3	Custom (Storage)
Network Adapter 4	Custom (VMotion)
USB Controller	Present
Display	Auto detect

6.2 EVE-NG mapping principle

In EVE-NG, Cloud/Pnet objects bridge an external Workstation network into the EVE lab. The Cloud object does not itself create a VLAN; the Cisco switch ports define access/trunk behavior. This keeps 802.1Q semantics identical to the intended Cisco design.

Example: logical trunk/access behavior; interface numbering follows the final IOL topology.

```
Core -> Access
interface Ethernet0/0
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30

Access -> Client
interface Ethernet0/1
 switchport mode access
 switchport access vlan 10
```

7. EVE-NG Cisco Network Topology

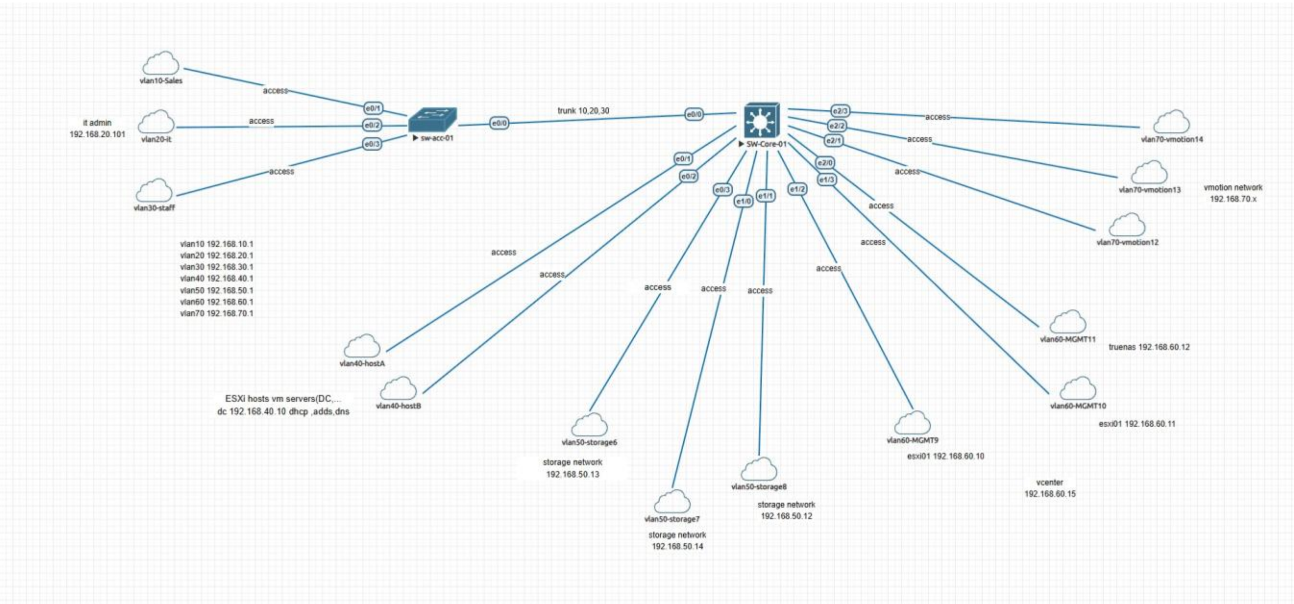


Figure 3 - Current EVE-NG topology screenshot captured during implementation

The current topology uses one Layer-2 access switch and one Layer-3 core device. The access switch carries VLAN 10/20/30 on a single 802.1Q trunk to the core. The core provides SVIs for VLAN 10/20/30/40/50/60/70.

7.1 Final stable IOL image used for the access-switch path

```
i86bi_linux_l2-adventerprisek9-ms.SSA.high_iron_20190423.bin
```

This image replaced an earlier IOSv-based path after repeatable CPU-hog / CPU exception behavior was observed during traffic tests.

7.2 Core L3 and Access L2 Ports and Interfaces

```
SW-Core-01
interface Ethernet0/0
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 switchport access vlan 40
 switchport mode access
!
interface Ethernet0/2
 switchport access vlan 40
 switchport mode access
!
interface Ethernet0/3
 switchport access vlan 50
 switchport mode access
```

```
SW-Acc-01
interface Ethernet0/0
 switchport trunk allowed vlan 10,20,30
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 switchport access vlan 10
 switchport mode access
!
interface Ethernet0/2
 switchport access vlan 20
 switchport mode access
!
interface Ethernet0/3
 switchport access vlan 30
  switchport mode access
 !
 interface Ethernet1/0
 !
```

```
!
interface Ethernet1/0
 switchport access vlan 50
 switchport mode access
!
interface Ethernet1/1
 switchport access vlan 50
 switchport mode access
!
interface Ethernet1/2
 switchport access vlan 60
 switchport mode access
!
interface Ethernet1/3
 switchport access vlan 60
 switchport mode access
!
interface Ethernet2/0
 switchport access vlan 60
 switchport mode access
!
interface Ethernet2/1
 switchport access vlan 70
 switchport mode access
!
interface Ethernet2/2
 switchport access vlan 70
 switchport mode access
!
interface Ethernet2/3
 switchport access vlan 70
 switchport mode access
!
interface Ethernet3/0
!
interface Ethernet3/1
!
interface Ethernet3/2
!
interface Ethernet3/3
!
```

8. Core L3 and Access L2 Design

8.1 Core role

- Layer-3 inter-VLAN routing via SVI gateways.
- DHCP relay using ip helper-address toward the DC at 192.168.40.10.
- Central point for user VLAN ACLs.
- STP root for the lab VLANs.
- Default route to a future Internet firewall (next phase).

8.2 Core SVI baseline

```
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 ip access-group ACL_SALES in
 ip helper-address 192.168.40.10
!
interface Vlan20
 ip address 192.168.20.1 255.255.255.0
 ip access-group ACL_IT in
 ip helper-address 192.168.40.10
!
interface Vlan30
 ip address 192.168.30.1 255.255.255.0
 ip access-group ACL_STAFF in
 ip helper-address 192.168.40.10
!
interface Vlan40
 ip address 192.168.40.1 255.255.255.0
!
interface Vlan50
 ip address 192.168.50.1 255.255.255.0
!
interface Vlan60
 ip address 192.168.60.1 255.255.255.0
!
interface Vlan70
 ip address 192.168.70.1 255.255.255.0
```

8.3 Core IP Route

```
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, Vlan10
L        192.168.10.1/32 is directly connected, Vlan10
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, Vlan20
L        192.168.20.1/32 is directly connected, Vlan20
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C        192.168.30.0/24 is directly connected, Vlan30
L        192.168.30.1/32 is directly connected, Vlan30
      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.40.0/24 is directly connected, Vlan40
L        192.168.40.1/32 is directly connected, Vlan40
      192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.50.0/24 is directly connected, Vlan50
L        192.168.50.1/32 is directly connected, Vlan50
      192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.60.0/24 is directly connected, Vlan60
L        192.168.60.1/32 is directly connected, Vlan60
      192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.70.0/24 is directly connected, Vlan70
L        192.168.70.1/32 is directly connected, Vlan70
```

8.3 Layer-2 management IP

The access switch remains Layer 2. Its management IP is assigned to an SVI in the IT VLAN, not to a physical port. Example used in the design:

```
interface Vlan20
 ip address 192.168.20.10 255.255.255.0
 no shutdown
!
ip default-gateway 192.168.20.1
```

The IP is for switch management only; VLAN20's default gateway remains 192.168.20.1 on the core.

9. Security Model and ACL Policy

The policy objective is intentionally simple: all three user VLANs have full IP access to the DC at 192.168.40.10 so DNS/DHCP/AD/file services can work without per-port service exceptions. The stronger restriction is placed on management, storage and vMotion networks.

Source	DC	Other user VLANs	Management	Storage / vMotion
Sales VLAN10	ALLOW	DENY (IT ALLOW)	DENY	DENY
IT VLAN20	ALLOW	ALLOW	Admin IP only	DENY
Staff VLAN30	ALLOW	DENY (IT ALLOW)	DENY	DENY
IT Admin .101	ALLOW	ALLOW	ALLOW	DENY*
VLAN40 Servers	As required	As required	Controlled	Storage allowed where required
VLAN50 Storage	ESXi only	DENY	Controlled	Isolated
VLAN70 vMotion	ESXi only	DENY	DENY	Isolated

*Management access is sourced from IT admin 192.168.20.101. Storage/vMotion remain restricted by design.

9.1 ACL design rule

Classic Cisco ACLs are stateless Because standard Cisco ACLs are stateless, return traffic must be explicitly permitted when the return path is evaluated by another inbound ACL. Stateful behavior would require a stateful firewall or a different ACL design.

```
ip access-list extended ACL_IT
 permit ip host 192.168.20.101 192.168.60.0 0.0.0.255      ! allow it admin access to mgmt
 deny   ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
 deny   ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny   ip 192.168.20.0 0.0.0.255 192.168.70.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_SALES
 permit ip 192.168.10.0 0.0.0.255 host 192.168.40.10       ! allow sales access to DC traffic
 deny   ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 deny   ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
 deny   ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny   ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
 deny   ip 192.168.10.0 0.0.0.255 192.168.70.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_STAFF
 permit ip 192.168.30.0 0.0.0.255 host 192.168.40.10       ! allow sales access to DC traffic
 deny   ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny   ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
 deny   ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny   ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255
 deny   ip 192.168.30.0 0.0.0.255 192.168.70.0 0.0.0.255
 permit ip any any
```

Direction

```
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 ip access-group ACL_SALES in           The ACL is applied inbound on the source vlan SVI so traffic
```

```
 ip helper-address 192.168.40.10
!
interface Vlan20
 ip address 192.168.20.1 255.255.255.0
 ip access-group ACL_IT in
 ip helper-address 192.168.40.10
!
interface Vlan30
 ip address 192.168.30.1 255.255.255.0
 ip access-group ACL_STAFF in
 ip helper-address 192.168.40.10
!
```

is filtered as it enters the layer 3 interface before being routed to other vlans.

9.2 Management source

The privileged IT admin workstation is 192.168.20.101. This address is the intended single source for switch SSH and access to ESXi, VCSA and TrueNAS management interfaces.

```
ip access-list standard SSH_ACCESS
 permit 192.168.20.101
 deny   any
!
```

10. Secure Device Management (SSH)

SSH is used instead of Telnet. The objective is to reduce the management plane to one explicitly authorized admin source address.

```
Hostname SW-Core-01
ip domain-name novincorp.local
username it.admin privilege 15 secret xxxxxxx

crypto key generate rsa 512
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

line vty 0 4
 login local
 transport input ssh
 access-class SSH_ACCESS in
```

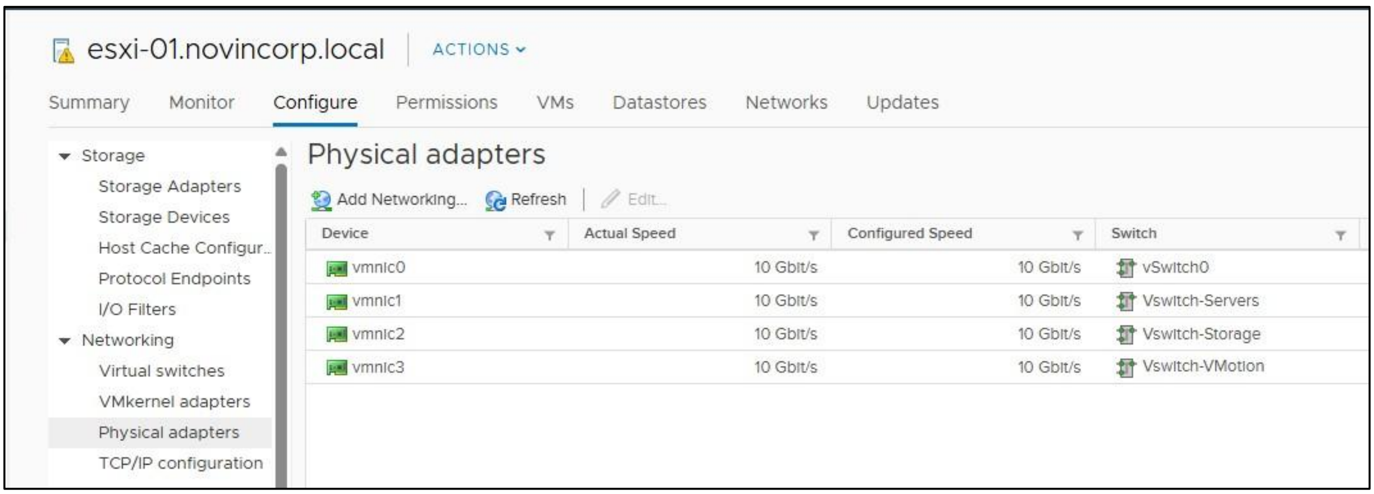
- Disable Telnet by allowing only SSH on the VTY lines.
- Do not enable HTTP management unless it is intentionally needed.
- Keep the management IP only on the management SVI (or controlled IT VLAN for the access-switch management IP).
- Record the final SSH verification output: show ip ssh and show users / show line.

11. ESXi Host Design

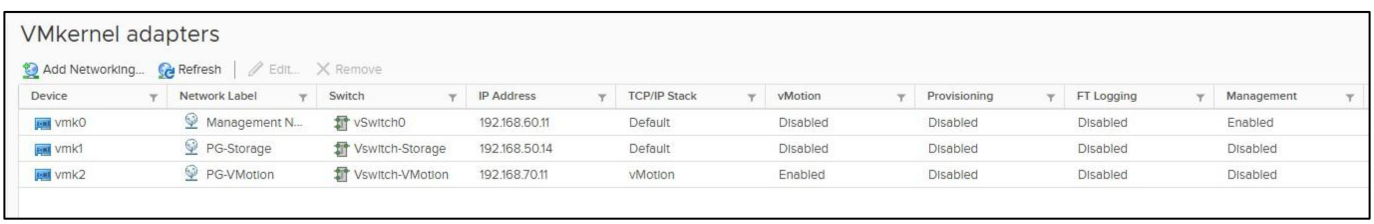
Uplink	Role	VLANs	Notes
vmnic0	Management	VLAN60	Dedicated management path
vmnic1	Server Data	VLAN40	vSwitch-Servers
Vmnic2	Storage	VLAN50	Dedicated storage path
Vmnic3	VMotion	VLAN70	Dedicated VMotion path
VMkernel	**Role**	**VLANS**	**Notes**
ESXI01 vmk0	ESXi management	VLAN60	Host management 192.168.60.10
ESXI02 vmk0	ESXi management	VLAN60	Host management 192.168.60.11
ESXI01 vmk1	iSCSI	VLAN50	Storage VMkernel 192.168.50.13
ESXI02 vmk1	iSCSI	VLAN50	Storage VMkernel 192.168.50.14
vmk2	vMotion	VLAN70	vMotion VMkernel when enabled

Vmnics and vmkernels both are the same in two ESXi host.

Vmnics in ESXi-01

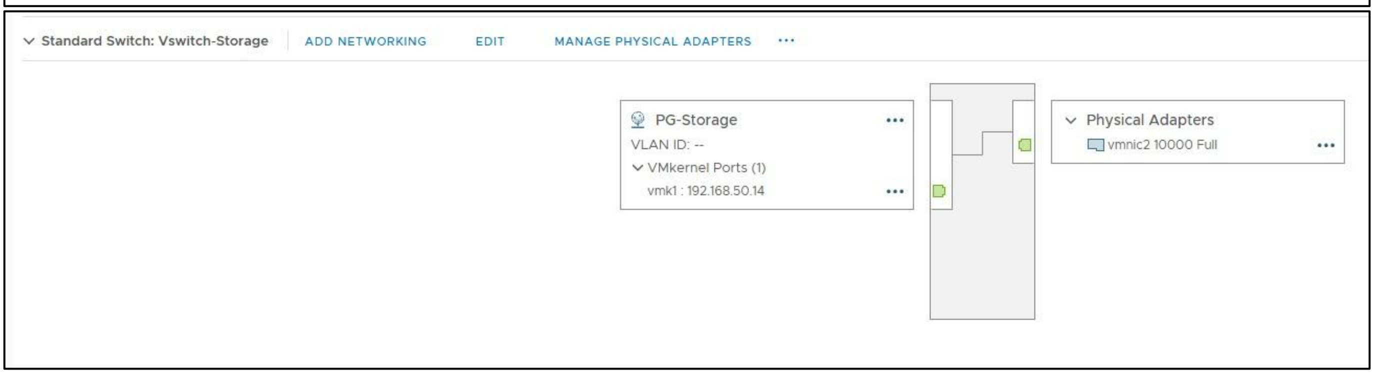


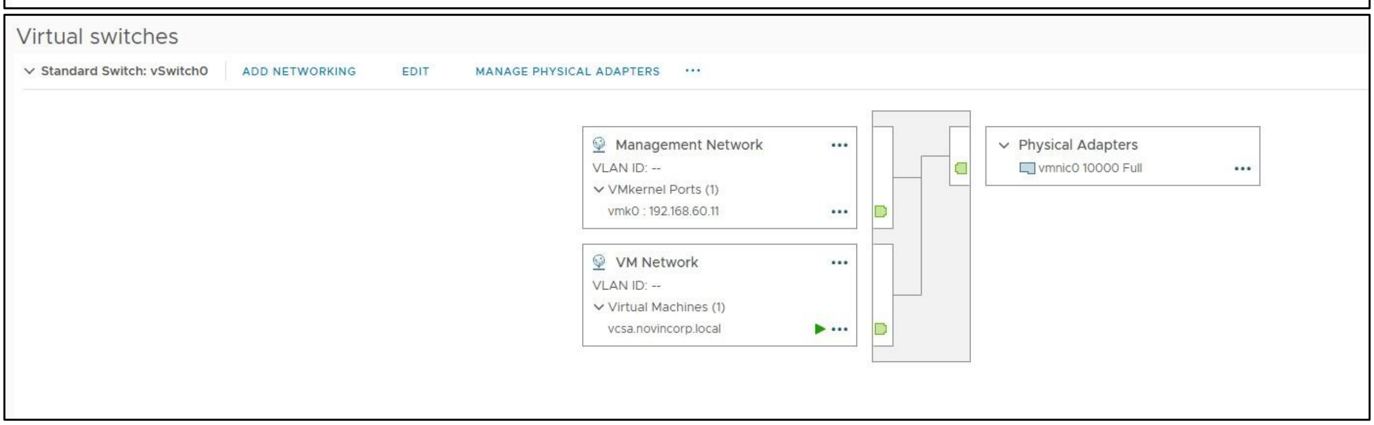
VMkernels in ESXi-02



11.1 Virtual switch intent

- Management uses a dedicated access network, preventing management from depending on the data trunk.
- Server VMs use a Server VLAN port group.
- Storage uses a dedicated Storage port group and VMkernel.
- vMotion uses a dedicated VMkernel on VLAN70.
- Standard vSwitches were retained for compatibility and simplicity in the lab.





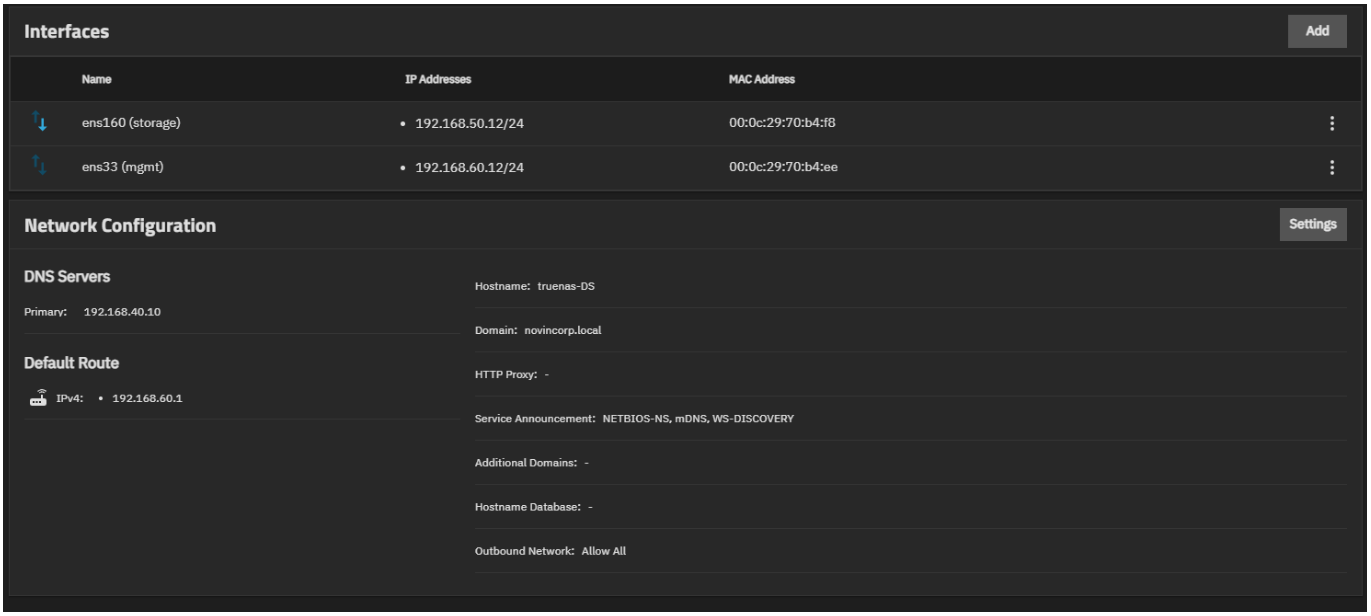
11.2 Nested Workstation trunk note

No VLAN trunk is used on the final ESXi uplink design. Management, Server, Storage and vMotion traffic use dedicated virtual NICs and corresponding Workstation VMnets.

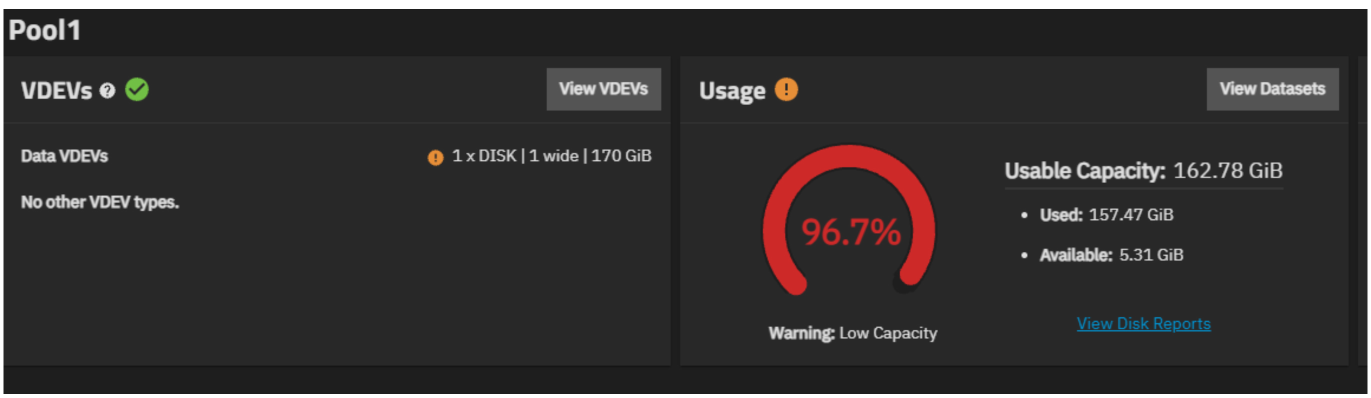
12. TrueNAS and Shared iSCSI Storage

- TrueNAS Management NIC: 192.168.60.12 on VLAN60.
- TrueNAS Storage NIC: dedicated storage network 192.168.50.12 on VLAN50.
- A Software Adaptor was added for iSCSI Adaptor
- Network Port Binding connected the iSCSI Adaptor and Dedicated VMKernel to each other.
- Data pool uses the lab-provisioned storage disk (approximately 170 GB allocated to the storage workload).
- A zvol is exposed through an iSCSI device extent and presented to the ESXi hosts.
- The shared datastore is used for VCSA and workload VMs so HA can restart VMs on the surviving ESXi host.

12.1 TrueNAS Networking



12.2 TrueNAS POOl & ZVol



The storage pool was intentionally sized tightly to fit the nested lab resource budget; the high utilization shown in the screenshot is a lab-capacity constraint, not a production sizing target.

Dataset	Used / Available	Encryption	Usage
Pool1	157.47 GiB / 5.31 GiB	Unencrypted	
ZVol01	157.43 GiB / 142.08 GiB	Unencrypted	iSCSI

12.3 ESXi iSCSI Adapter and Storage Devices

Storage Adapters

Add Software Adapter | Refresh | Rescan Storage... | Rescan Adapter

Adapter	Type	Status	Identifier
Model: iSCSI Software Adapter			
vmhba65	iSCSI	Online	iqn.1998-01.com.vmware:6a7cbbaf-e48d-405d-6fea-000c29245a59-5ab6d085
Model: PIIX4 for 430TX/440BX/MX IDE Controller			
vmhba1	Block SCSI	Unknown	--
vmhba64	Block SCSI	Unknown	--
Model: PVSCSI SCSI Controller			
vmhba0	SCSI	Unknown	--

Storage Devices

Refresh | Attach | Detach | Rename...

Name	LUN	Type	Capacity	Datastore	Operational State
Local VMware, Disk (mpx.vmhba0:C0:T0:L0)	0	disk	10.00 GB	Local-DS-01	Attached
Local NECVMWar CD-ROM (mpx.vmhba64:C0:T0:L0)	0	cdrom		Not Consumed	Attached
TrueNAS iSCSI Disk (naa.6589cfc000000b6ca41b52ebbfb6e75b)	0	disk	155.00 GB	iSCSI-DS	Attached

12.4 ESXi-side validation commands

Engineering lesson: iSCSI datastore creation initially hung when the storage path traversed the earlier GNS3 virtual switching layer. When the iSCSI path was tested directly on a dedicated VMnet, datastore creation succeeded. This isolated the problem to the network emulation path rather than the TrueNAS zvol itself.

```
[root@ESXi-01:~] esxcli iscsi adapter list
Adapter  Driver     State   UID                                                                   Description
-------  ---------  ------  --------------------------------------------------------------------  ---------------------
vmhba65  iscsi_vmk  online  iqn.1998-01.com.vmware:6a7cbbaf-e48d-405d-6fea-000c29245a59-5ab6d085  iSCSI Software Adapter
```

```
[root@ESXi-01:~] esxcli storage filesystem list
Mount Point                                        Volume Name  UUID                                 Mounted  Type            Size          Free
-------------------------------------------------  -----------  -----------------------------------  -------  ------  ------------  ------------
/vmfs/volumes/6a7cbc92-930f06da-305c-000c29245a59  Local-DS-01  6a7cbc92-930f06da-305c-000c29245a59     true  VMFS-6    2684354560    1175453696
/vmfs/volumes/6a7f9a80-13630744-fc53-000c29245a59  iSCSI-DS     6a7f9a80-13630744-fc53-000c29245a59     true  VMFS-6  166161547264  115958874112
/vmfs/volumes/6a7cbc8c-88fe0b62-3d87-000c29245a59               6a7cbc8c-88fe0b62-3d87-000c29245a59     true  vfat       299712512     118415360
/vmfs/volumes/6a7cbc92-a5bd14d1-b678-000c29245a59               6a7cbc92-a5bd14d1-b678-000c29245a59     true  vfat      4293591040    4268163072
/vmfs/volumes/4b531d7f-ac8718d7-67ab-332cdb61b310               4b531d7f-ac8718d7-67ab-332cdb61b310     true  vfat       261853184      99147776
/vmfs/volumes/3eff6370-680afec5-fc52-63553442d433               3eff6370-680afec5-fc52-63553442d433     true  vfat       261853184     108228608
```

13. Active Directory, DNS, DHCP

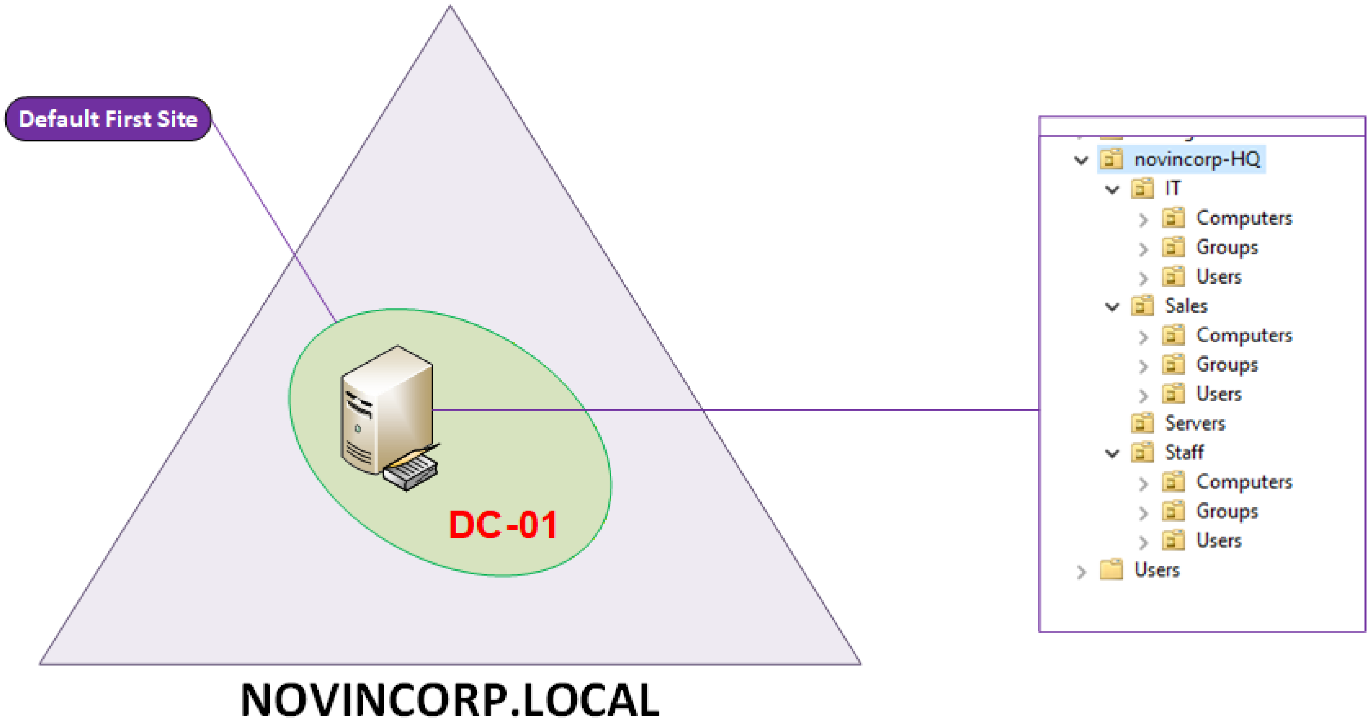


Figure 4 - Active Directory organizational structure captured during implementation

The AD structure is organized by department and role. The lab uses Global groups for user membership and Domain Local groups for resource permissions, following the AGDLP pattern.

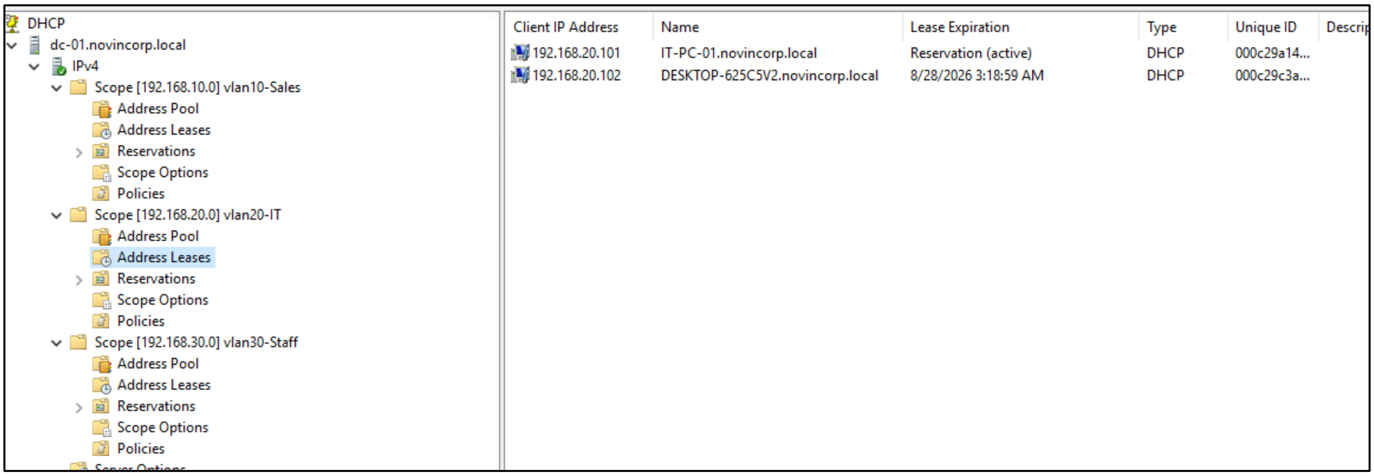
13.1 Department group model

User -> Global Group -> Domain Local Group -> Resource ACL

DL-IT	Group
Global-IT	Group
DL-Sales	Group
Global-Sales	Group
DL-Staff	Group
Global-Staff	Group

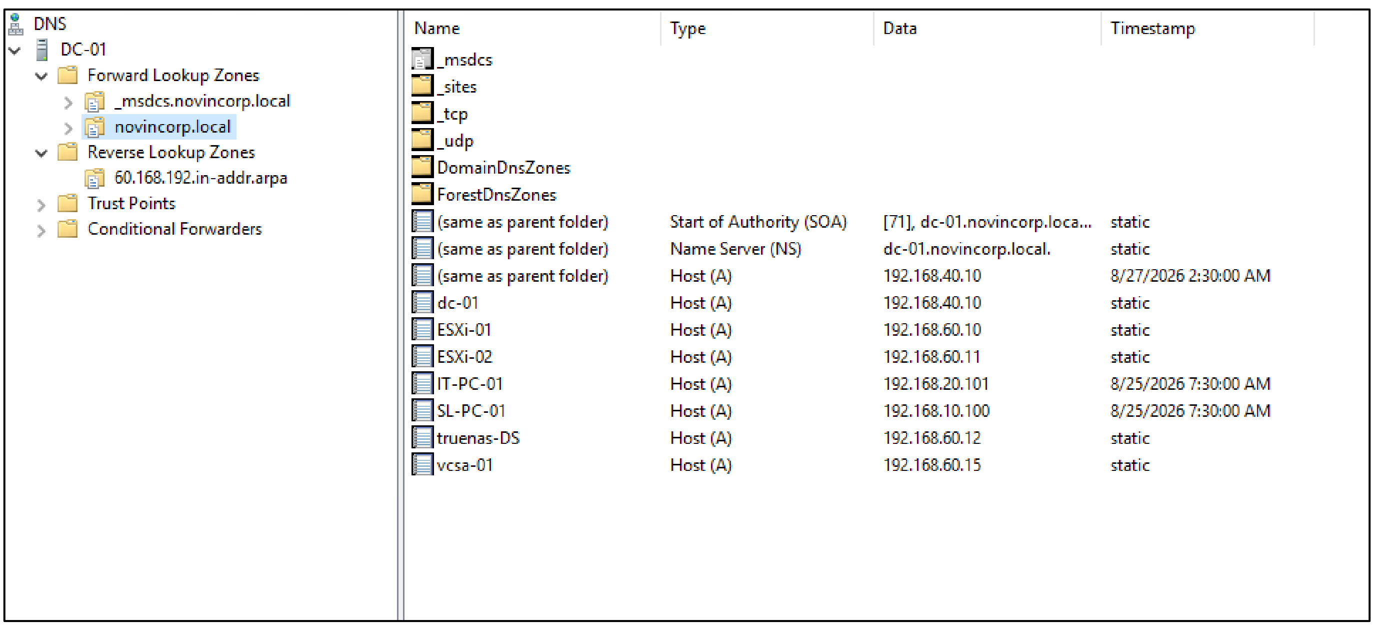
13.2 DHCP relay

The DC is the only DHCP server. The core forwards DHCP broadcasts from VLAN10/20/30 using ip helper-address 192.168.40.10. This keeps DHCP centralized while preserving VLAN isolation.



13.3 DNS records

- Forward records exist for ESXi01 (192.168.60.10), ESXi02 (192.168.60.11), TrueNAS (192.168.60.12), VCSA (192.168.60.15) and DC (192.168.40.10).
- Reverse DNS for management services uses the 192.168.60.0/24 reverse zone.
- VCSA deployment depends on working forward and reverse DNS for its FQDN.



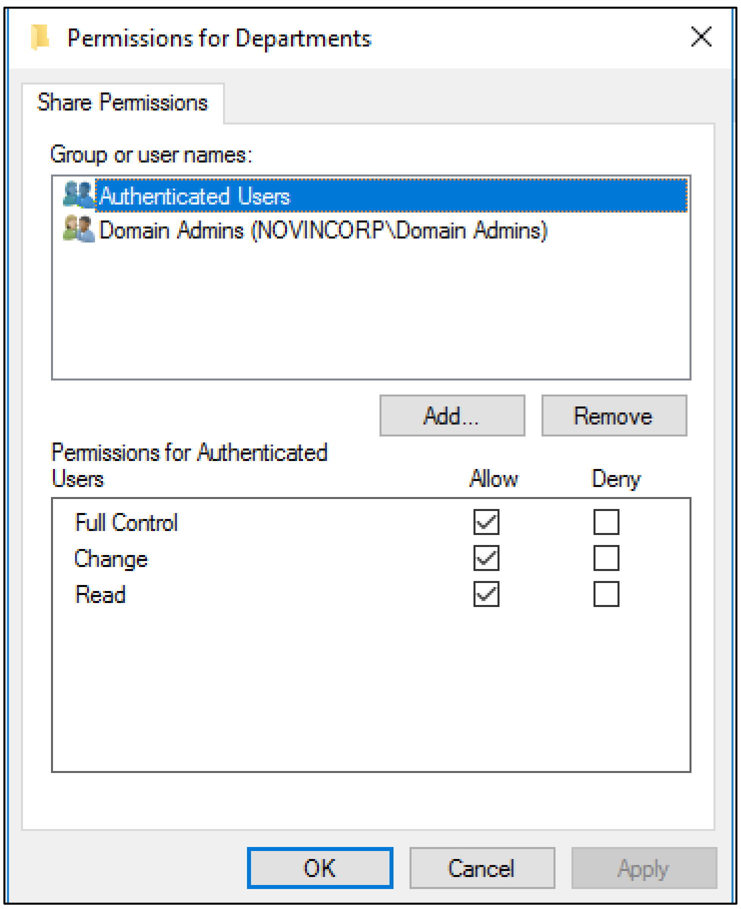
14. File Server, NTFS/SMB Permissions and GPOs

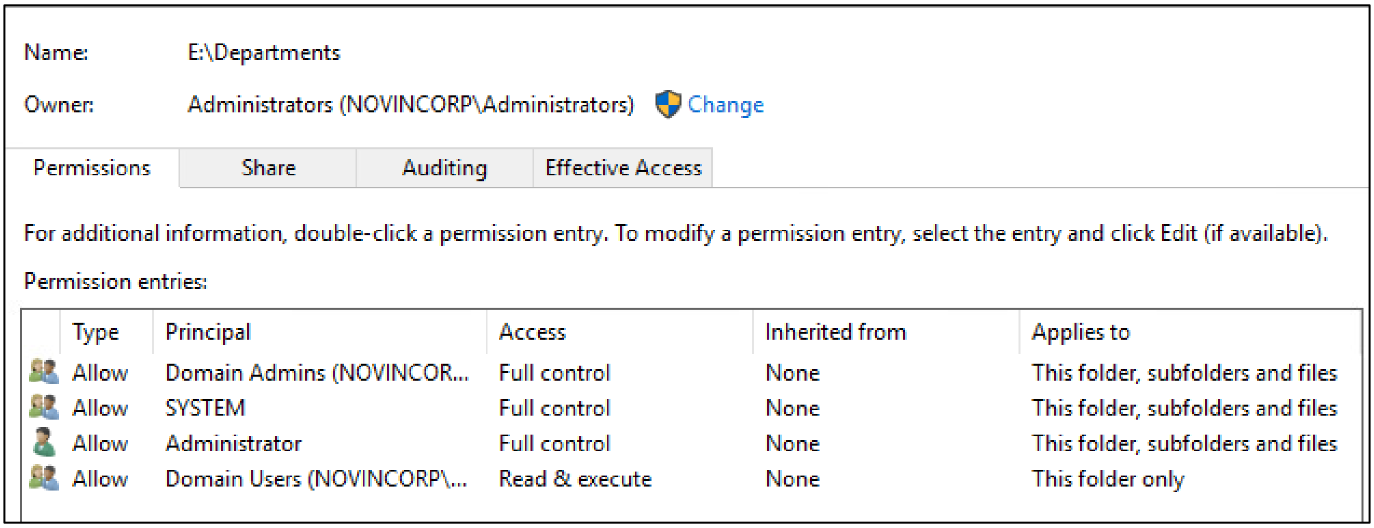
```
E:\
└── Departments
    ├── IT
    ├── Sales
    ├── Staff
    └── Public
```

The parent folder is shared as a single SMB share (for example \\DC\Departments). NTFS ACLs provide the real security boundary. Share permissions are intentionally simple to make the effective access model easier to audit.

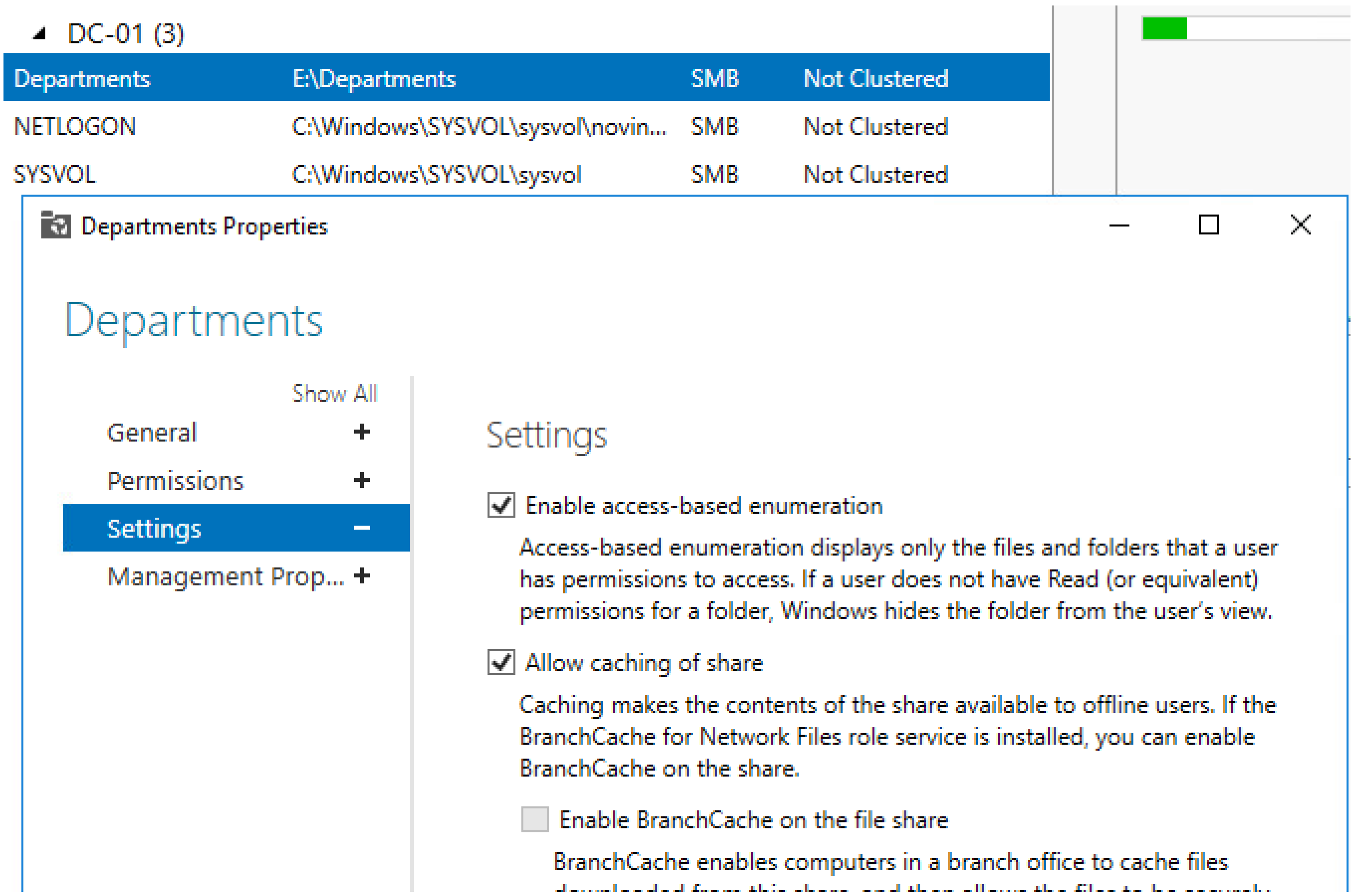
Access-based enumeration has been enabled and displays folders to users only have at least read permissions.

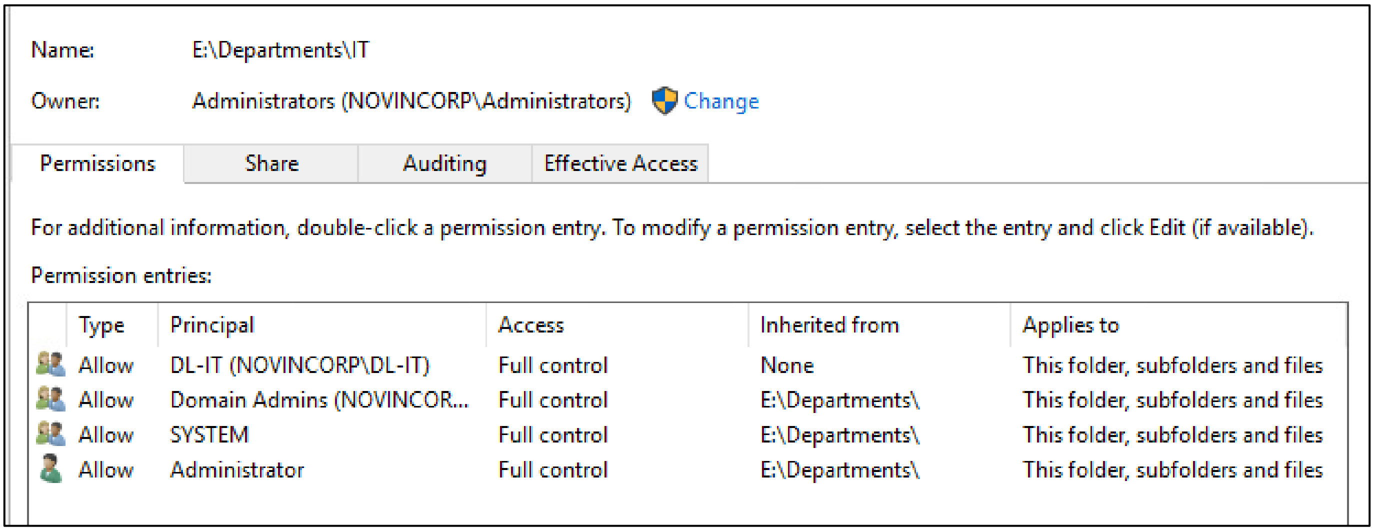
Folder	Department group	User access	Admin access
IT	DL-IT	Full/Modify for IT only	Domain Admins / SYSTEM
Sales	DL-Sales	Modify for Sales only	Domain Admins / SYSTEM
Staff	DL-Staff	Modify for Staff only	Domain Admins / SYSTEM
Public	Staff+sales+IT + Creator Owner	Read/Create; user controls own created files	IT + Domain Admins Full

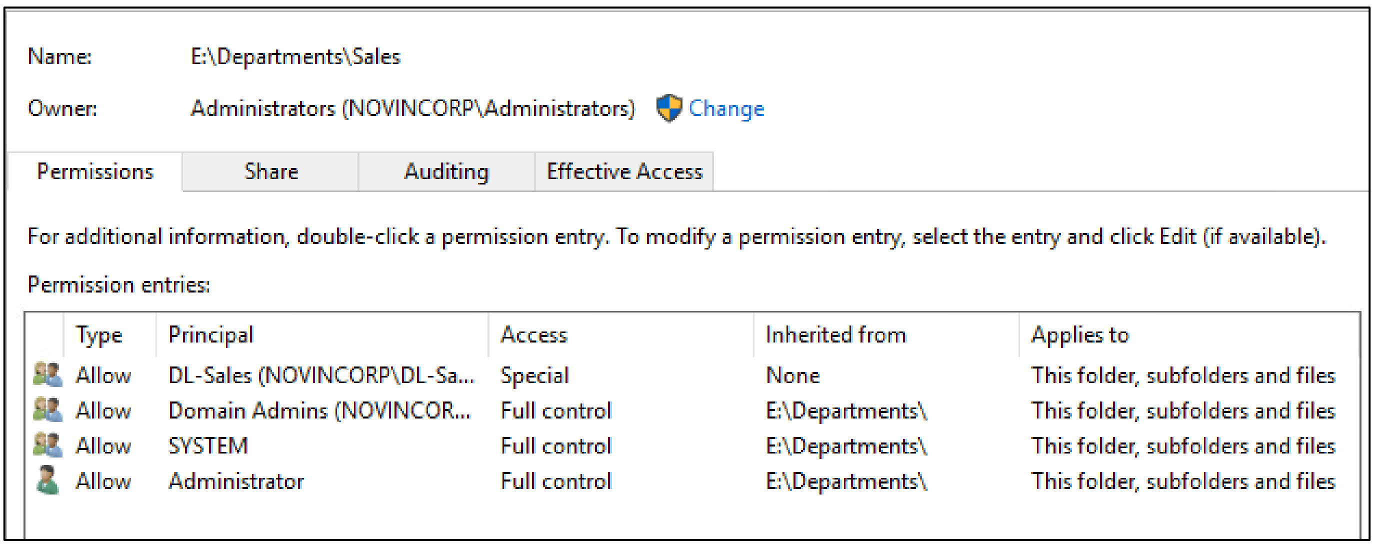




Access-based enumeration enabled on Departments







14.1 Public folder design

The Public requirement cannot be implemented by granting Modify to all users because Modify includes Delete. The lab instead uses controlled Create/Read permissions for Domain Users and Creator Owner permissions so users can control files they create without receiving delete rights over other users' files.

Name: E:\Departments\Public

Owner: Administrators (NOVINCORP\Administrators) Change

Permissions | Share | Auditing | Effective Access

For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).

Permission entries:

Type	Principal	Access	Inherited from	Applies to
Allow	DL-IT (NOVINCORP\DL-IT)	Full control	None	This folder, subfolders and files
Allow	DL-Staff (NOVINCORP\DL-Sta...	Read, write & execute	None	This folder, subfolders and files
Allow	DL-Sales (NOVINCORP\DL-Sa...	Read, write & execute	None	This folder, subfolders and files
Allow	Administrators (NOVINCORP\...	Special	None	This folder only
Allow	CREATOR OWNER	Special	None	Subfolders and files only
Allow	Domain Admins (NOVINCOR...	Full control	E:\Departments\	This folder, subfolders and files
Allow	SYSTEM	Full control	E:\Departments\	This folder, subfolders and files
Allow	Administrator	Full control	E:\Departments\	This folder, subfolders and files

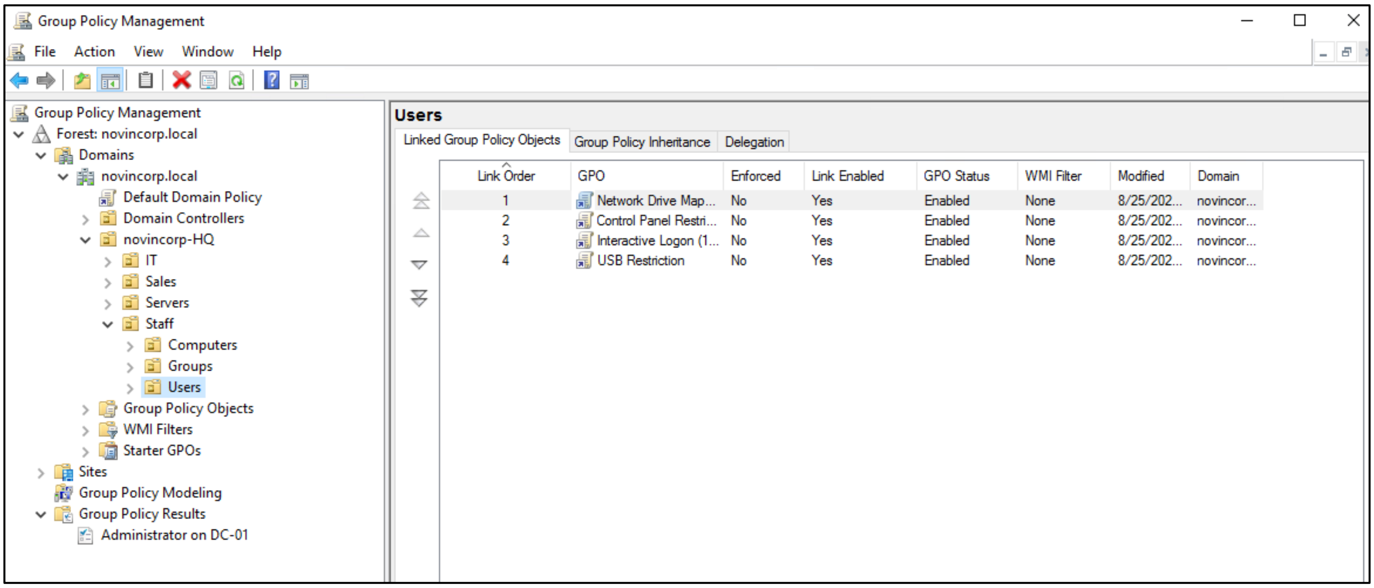
14.2 GPO evidence

GPO name	Scope	Description
Default Domain Policy	Novincorp.local	Length=7, complex, threshold lockdown=5, lock=5 min, ages=90
Map Drive	OU-IT, Sales, Staff	Department folder mapping for users
Control Panel Restriction	OU-Sales, Staff	Disabling control panel
USB restriction	OU-Sales, Staff	Restricts USB and removable storage devices
Screen Lock	OU-IT, Sales, Staff	Lockdown mode after 10 min inactivity

Group Policy Objects in novincorp.local

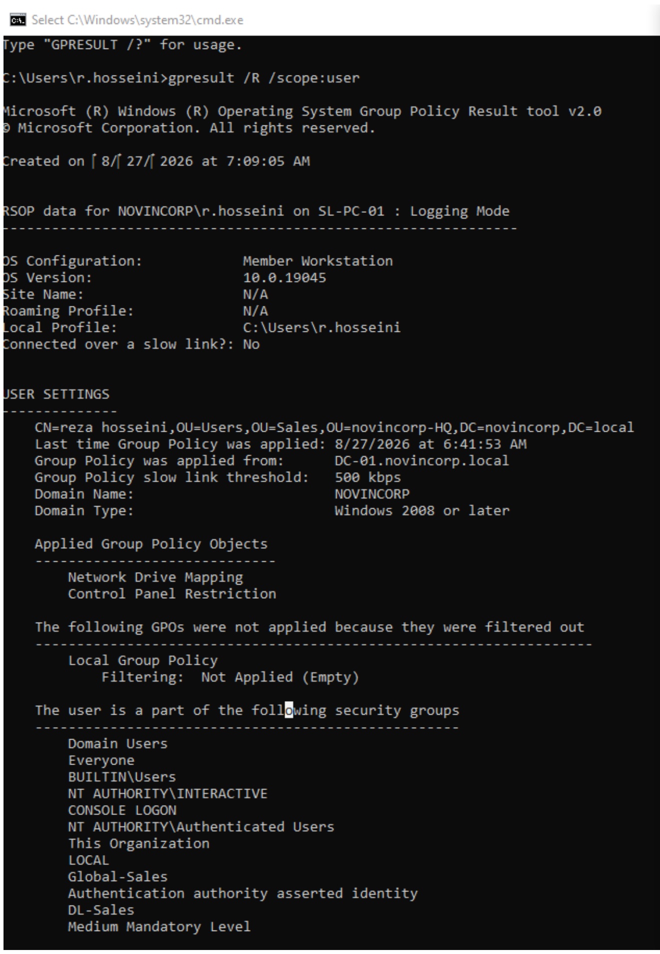
Contents | Delegation

Name	GPO Status	WMI Filter	Modified	Owner
Control Panel Restriction	Enabled	None	8/25/2026 9:01:02 AM	Domain Admins (N...
Default Domain Controllers Policy	Enabled	None	8/15/2026 11:15:06 ...	Domain Admins (N...
Default Domain Policy	Enabled	None	8/25/2026 8:37:04 AM	Domain Admins (N...
Interactive Logon (10min)	Enabled	None	8/25/2026 9:16:22 AM	Domain Admins (N...
Network Drive Mapping	Enabled	None	8/25/2026 8:25:58 AM	Domain Admins (N...
USB Restriction	Enabled	None	8/25/2026 9:14:06 AM	Domain Admins (N...

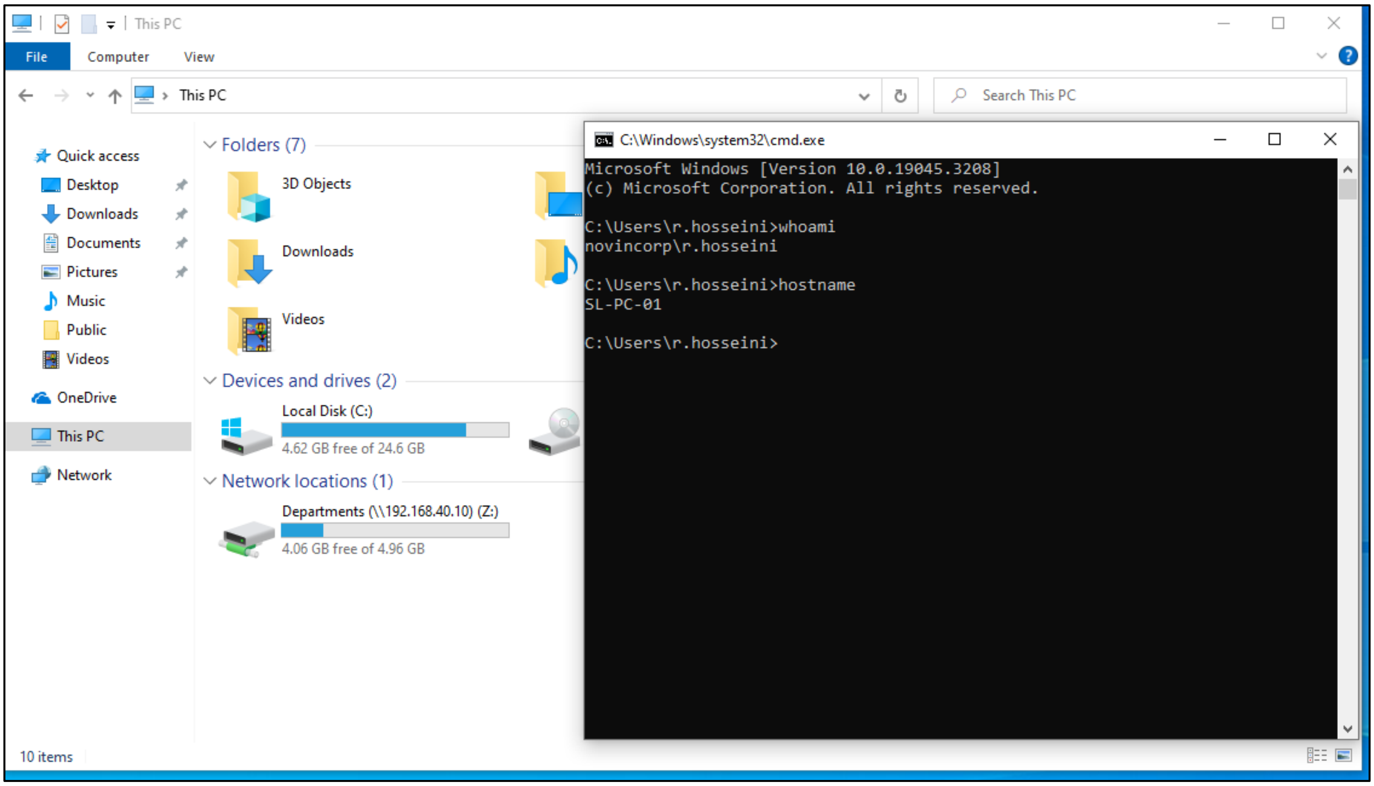


Gpresult /R /Scope: user

gpresult /R /scope: computer



Network Drive Mapping GPO on OU = sales users final result


This PC
File
Computer
View
This PC
Search This PC
Quick access
Desktop
Downloads
Documents
Pictures
Music
Public
Videos
OneDrive
This PC
Network
Folders (7)
3D Objects
Downloads
Videos
Devices and drives (2)
Local Disk (C:)
4.62 GB free of 24.6 GB
Network locations (1)
Departments (\\192.168.40.10) (Z:)
4.06 GB free of 4.96 GB
10 items
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3208]
(c) Microsoft Corporation. All rights reserved.
C:\Users\r.hosseini>whoami
novincorp\r.hosseini
C:\Users\r.hosseini>hostname
SL-PC-01
C:\Users\r.hosseini>

15. vCenter, HA and Virtualization Feature Validation

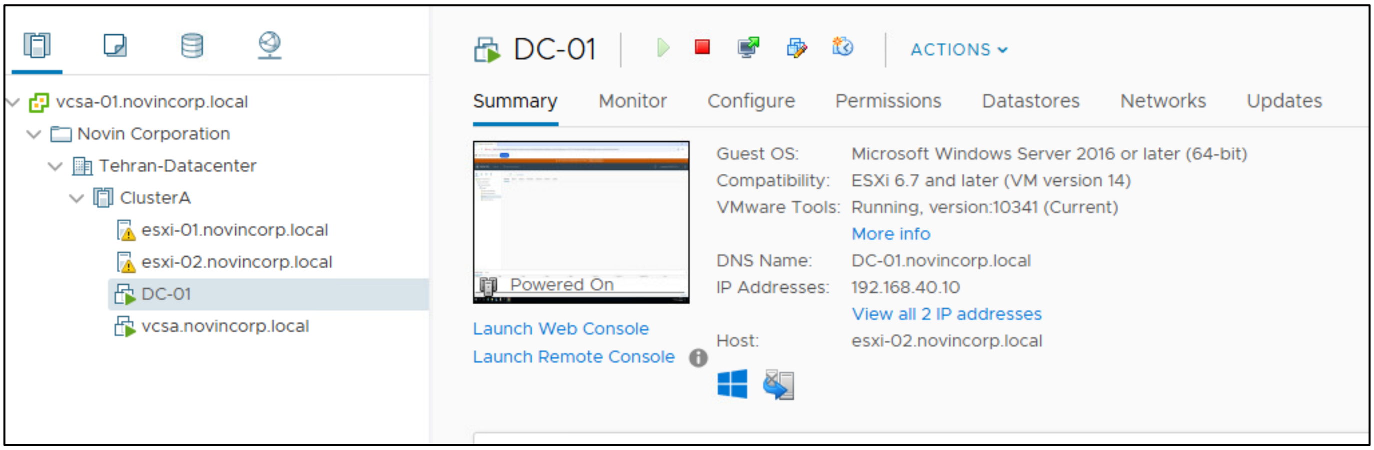
VCSA is a VM placed on the shared iSCSI datastore so its files are not pinned to a single local ESXi datastore. VCSA uses Management VLAN 60 (192.168.60.15).

15.1 HA

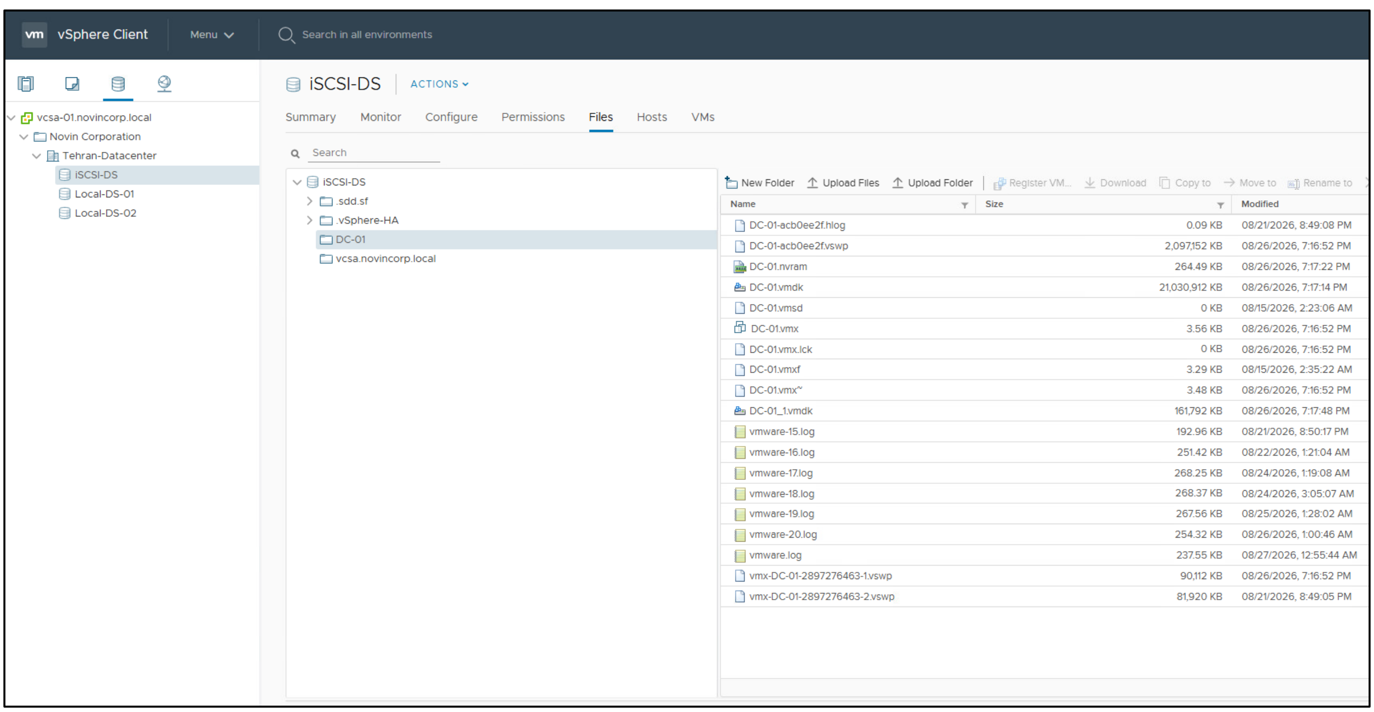
HA was tested in the lab: a host failure scenario was used to validate that the shared-datastore VMs had been restarted on the surviving ESXi host. Fault Tolerance was intentionally not used because the lab resource budget is too constrained for a meaningful FT demonstration.

15.2 HA Test Evidence

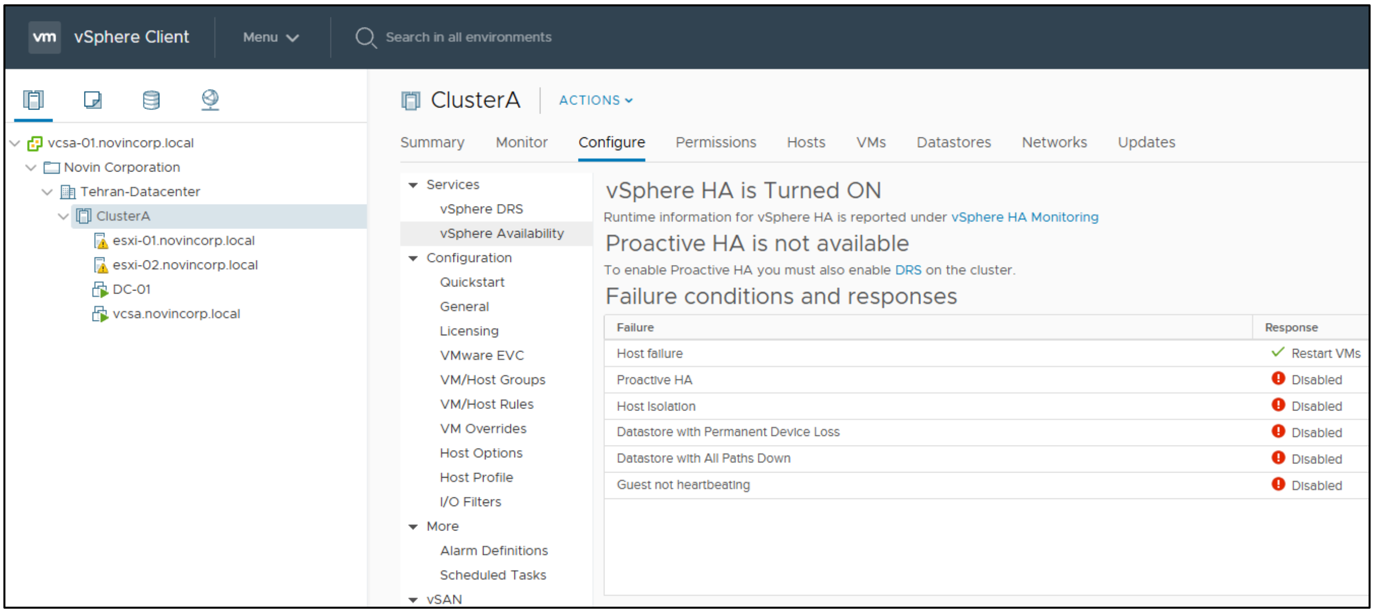
Two vms are running on esxi02



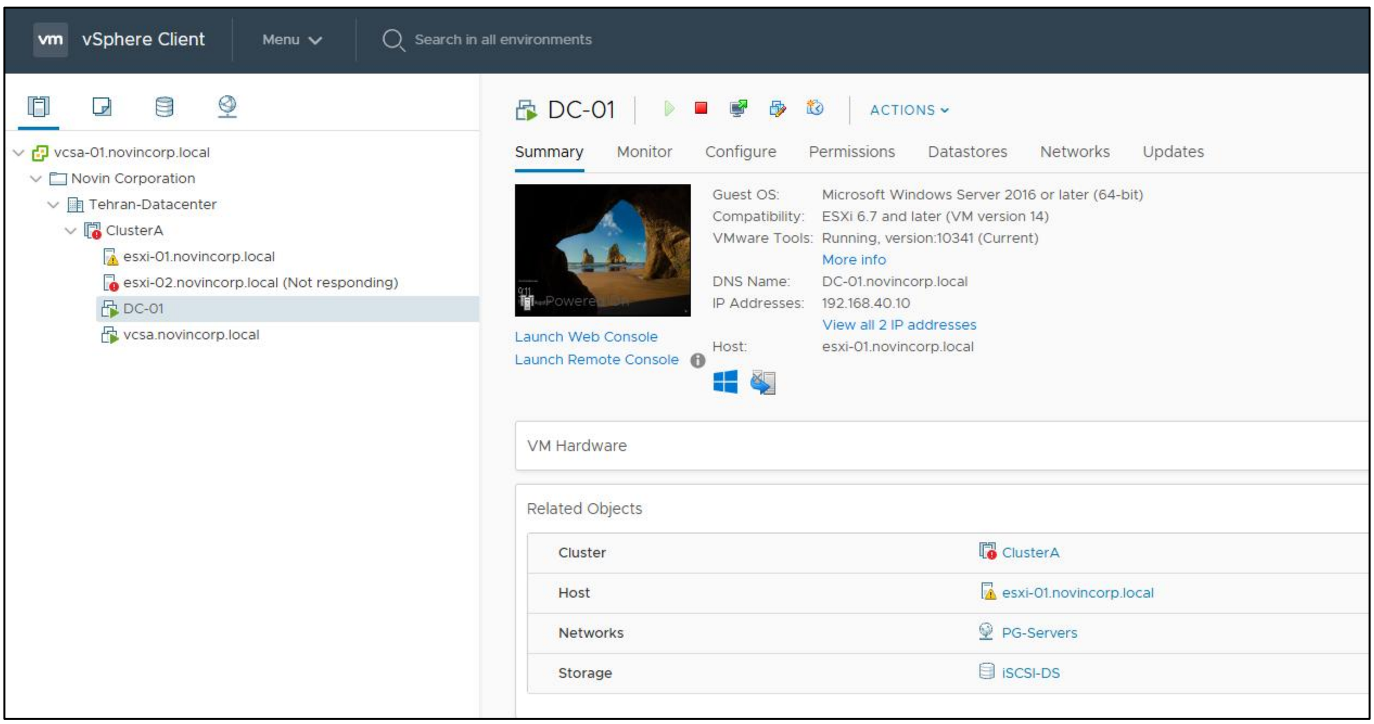
Both VMs use iSCSI Shared Storage



HA is enabled in cluster with restarting VMs and default setting



HA detected the host failure and restarted the affected VMs on the surviving ESXi host using the shared datastore.



C:\Windows\system32\cmd.exe

```
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
```

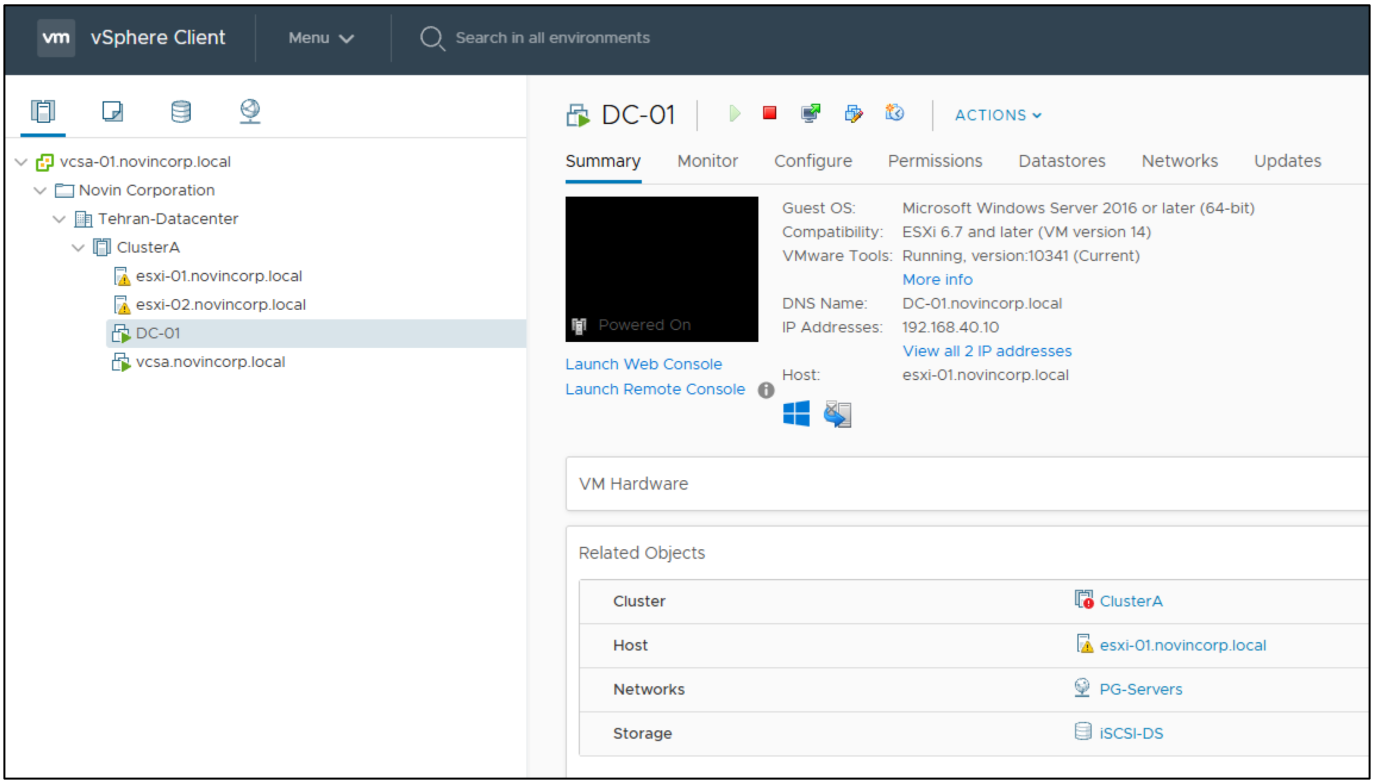
Vcenter event log

Insufficient resources to satisfy vSphere HA failover level on cluster ClusterA in Tehran-Datacenter	Error	08/27/2026, 1:08:10 AM		ClusterA		vim.event.InsufficientFailov...
Task: Deploy plug-in	Information	08/27/2026, 1:08:09 AM	Deploy plug-in	vcsa-01.novincorp.local	VSPHERE.LOCA...	vim.event.TaskEvent
Task: Deploy plug-in	Information	08/27/2026, 1:08:09 AM	Deploy plug-in	vcsa-01.novincorp.local	VSPHERE.LOCA...	vim.event.TaskEvent
User VSPHERE.LOCAL\vsphere-webclient-c97731bc-50a5-4e18-b533-8df8c9e43b63@127.0.0.1 logged in as h5-client/6.5.0	Information	08/27/2026, 1:08:06 AM			VSPHERE.LOCA...	vim.event.UserLoginSession...
Task: Deploy plug-in	Information	08/27/2026, 1:08:05 AM	Deploy plug-in	vcsa-01.novincorp.local	VSPHERE.LOCA...	vim.event.TaskEvent
Task: Check new notifications	Information	08/27/2026, 1:08:02 AM	Check new no...	vcsa-01.novincorp.local	com.vmware.vcl...	vim.event.TaskEvent
Task: Download patch definitions	Information	08/27/2026, 1:08:02 AM	Download pat...	vcsa-01.novincorp.local	com.vmware.vcl...	vim.event.TaskEvent
Alarm 'vSphere HA failover in progress' on ClusterA changed from Yellow to Green	Information	08/27/2026, 1:08:01 AM		ClusterA		vim.event.AlarmStatusChan...
Alarm 'vSphere HA failover in progress' on ClusterA changed from Green to Yellow	Information	08/27/2026, 1:08:01 AM		ClusterA		vim.event.AlarmStatusChan...
Alarm 'vSphere HA host status' on esxi-02.novincorp.local changed from Green to Red	Information	08/27/2026, 1:08:01 AM		esxi-02.novincorp.local		vim.event.AlarmStatusChan...
vSphere HA detected a possible host failure of host esxi-02.novincorp.local in cluster ClusterA in datacenter Tehran-Datacenter	Error	08/27/2026, 1:08:00 AM		esxi-02.novincorp.local		com.vmware.vc.HA.DasHost...
vSphere HA completed a virtual machine failover action in cluster ClusterA in datacenter Tehran-Datacenter	Information	08/27/2026, 1:06:28 AM		ClusterA		com.vmware.vc.HA.ClusterF...
vSphere HA restarted virtual machine DC-01 on host esxi-01.novincorp.local in cluster ClusterA	Warning	08/27/2026, 1:05:32 AM		DC-01		com.vmware.vc.ha.VmRestar...
vSphere HA restarted virtual machine vcsa.novincorp.local on host esxi-01.novincorp.local in cluster ClusterA	Warning	08/27/2026, 1:05:31 AM		vcsa.novincorp.local		com.vmware.vc.ha.VmRestar...
vSphere HA initiated a failover action on 2 virtual machines in cluster ClusterA in datacenter Tehran-Datacenter	Warning	08/27/2026, 1:05:31 AM		ClusterA		com.vmware.vc.HA.ClusterF...

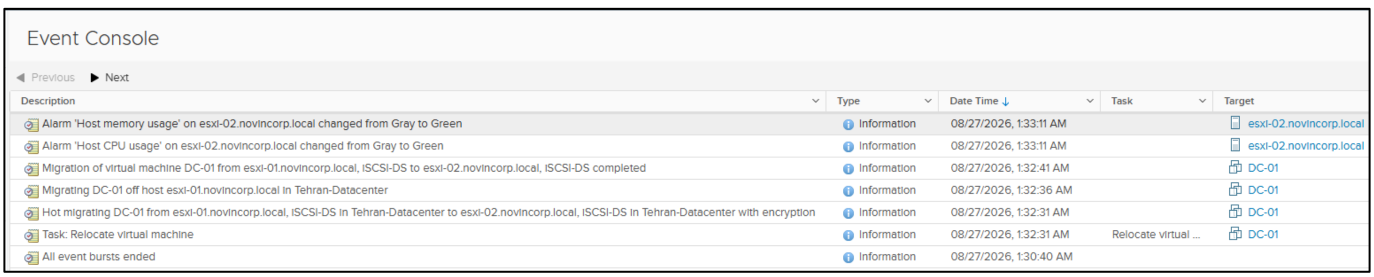
15.2 vMotion

- vMotion is designed as a dedicated VLAN70 VMkernel service.
- The VM was migrated between ESXi hosts while its VMDKs remained on the shared iSCSI datastore. In this nested lab, a brief network interruption was observed during some tests; the guest operating system was not intentionally rebooted by vMotion.

Vm (DC-01) is on esxi01



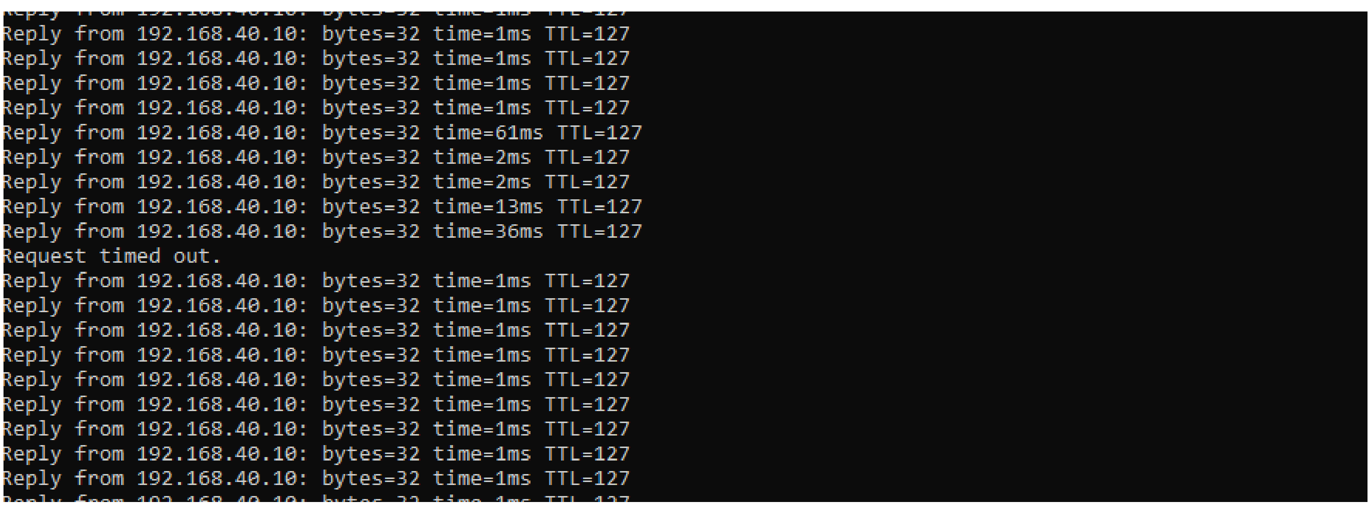
Event log after migrating is finished and pinging of the vm during the vmotion process



Event Console

Previous Next

Description	Type	Date Time	Task	Target
Alarm 'Host memory usage' on esxi-02.novincorp.local changed from Gray to Green	Information	08/27/2026, 1:33:11 AM		esxi-02.novincorp.local
Alarm 'Host CPU usage' on esxi-02.novincorp.local changed from Gray to Green	Information	08/27/2026, 1:33:11 AM		esxi-02.novincorp.local
Migration of virtual machine DC-01 from esxi-01.novincorp.local, iSCSI-DS to esxi-02.novincorp.local, iSCSI-DS completed	Information	08/27/2026, 1:32:41 AM		DC-01
Migrating DC-01 off host esxi-01.novincorp.local in Tehran-Datacenter	Information	08/27/2026, 1:32:36 AM		DC-01
Hot migrating DC-01 from esxi-01.novincorp.local, iSCSI-DS in Tehran-Datacenter to esxi-02.novincorp.local, iSCSI-DS in Tehran-Datacenter with encryption	Information	08/27/2026, 1:32:31 AM		DC-01
Task: Relocate virtual machine	Information	08/27/2026, 1:32:31 AM	Relocate virtual ...	DC-01
All event bursts ended	Information	08/27/2026, 1:30:40 AM		



```
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=61ms TTL=127
Reply from 192.168.40.10: bytes=32 time=2ms TTL=127
Reply from 192.168.40.10: bytes=32 time=2ms TTL=127
Reply from 192.168.40.10: bytes=32 time=13ms TTL=127
Reply from 192.168.40.10: bytes=32 time=36ms TTL=127
Request timed out.
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
```

16. Validation and Test Plan

Test	Expected result	Status	Evidence
Sales DHCP	Lease from DC via helper	PASS	Screenshot
Sales -> DC	Full IP/service access	PASS	Ping + service
Sales # Staff	Denied by ACL policy	PASS	Ping/TCP
IT Admin -> Management VLAN	Allowed from 192.168.20.101	PASS	HTTPS/SSH
Other user -> Management	Denied	PASS	ACL counter
ESXi -> TrueNAS iSCSI	LUN/datastore visible	PASS	ESXi screenshot
VCSA -> ESXi hosts	Both hosts managed	PASS	VCSA screenshot
HA host failure	VM restarts on surviving host	PASS	VCSA Events
Shared file permissions	Department/Public behavior correct	PASS	User evidence

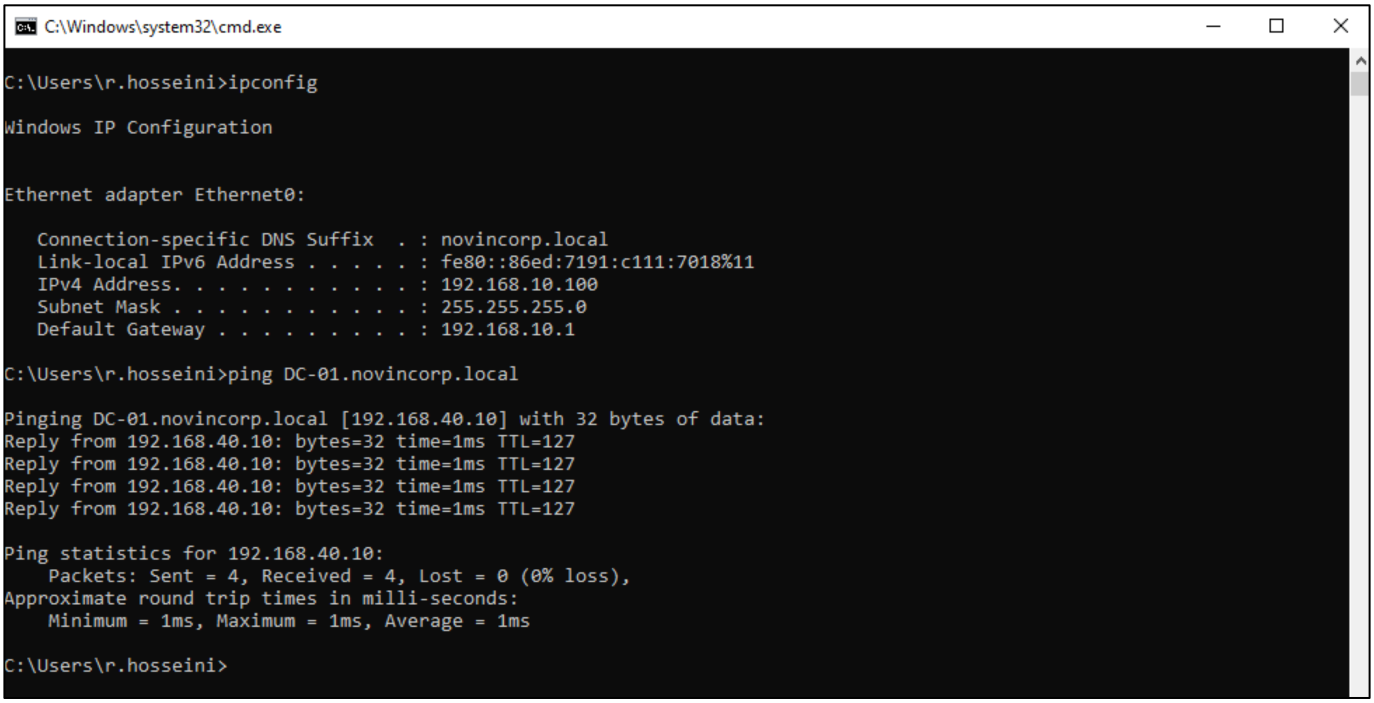
Sales obtain IP from DHCP

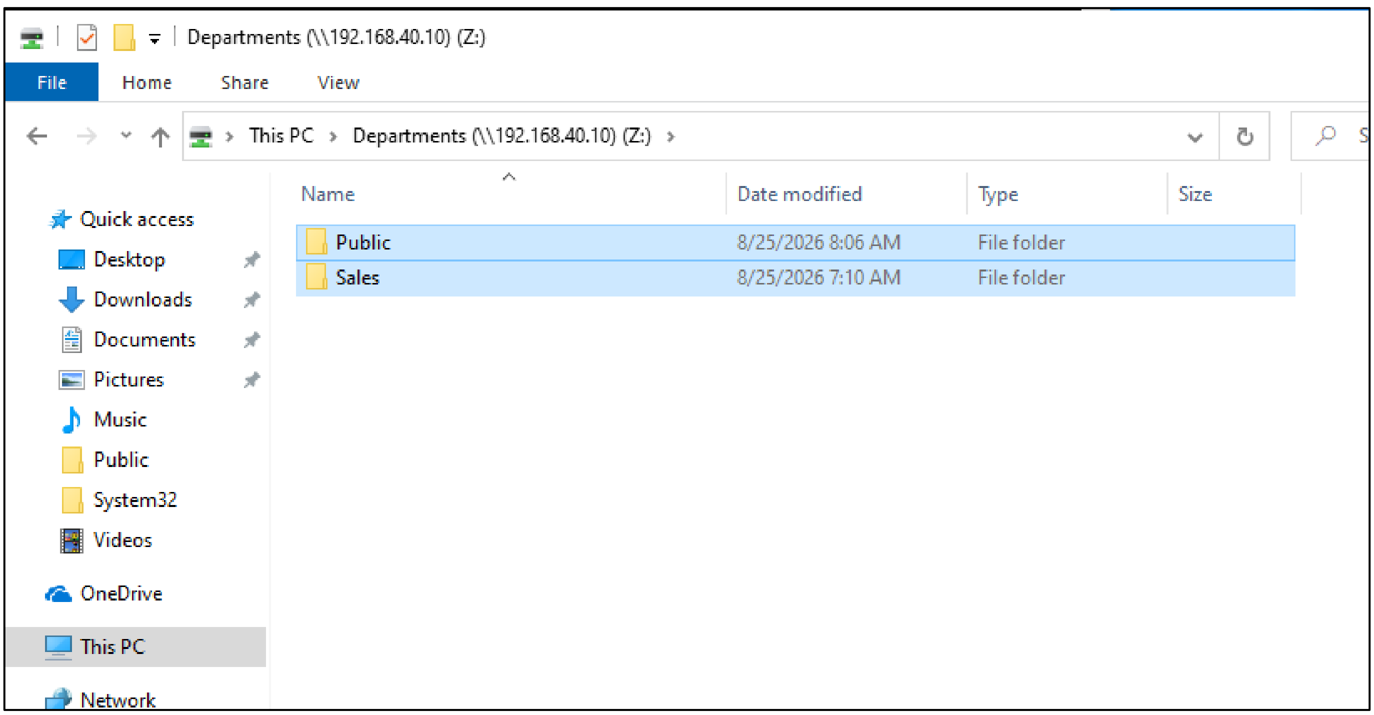
Client IP Address	Name	Lease Expiration	Type	Unique ID	Des...	Network A...	Prob...	Filter Profile	Policy
192.168.10.100	SL-PC-01.novincorp.lo...	9/5/2026 4:50:27 AM	DHCP	000c29c3ad7d		Full Access	N/A	None	

C:\Windows\system32\cmd.exe

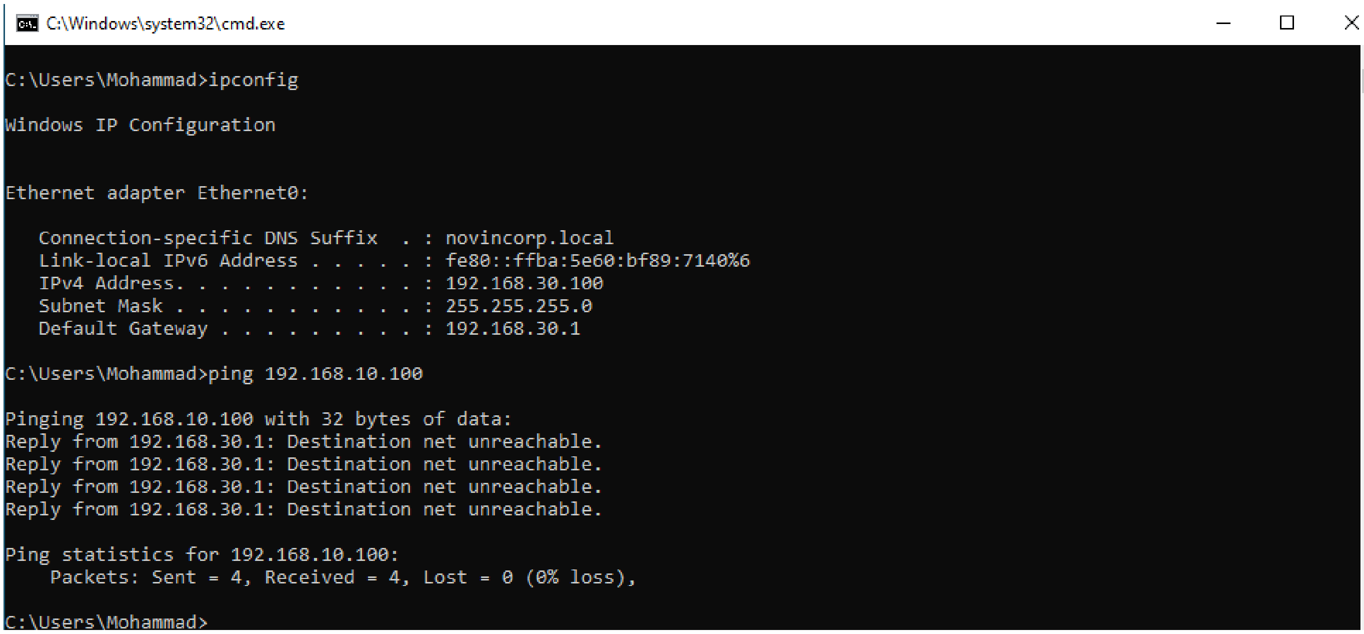
```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : SL-PC-01
   Primary Dns Suffix  . . . . . . . : novincorp.local
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : novincorp.local

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : novincorp.local
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-C3-AD-7D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::86ed:7191:c111:7018%11(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.100(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, August 28, 2026 3:50:22 AM
   Lease Expires . . . . . . . . . . : Saturday, September 5, 2026 3:50:21 AM
   Default Gateway . . . . . . . . . : 192.168.10.1
   DHCP Server . . . . . . . . . . . : 192.168.40.10
   DHCPv6 IAID . . . . . . . . . . . : 100666409
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-0D-37-CE-00-0C-29-C3-AD-7D
   DNS Servers . . . . . . . . . . . : 192.168.40.10
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\r.hosseini>_
```

Sales ping DC FQDN and access SMB shared file

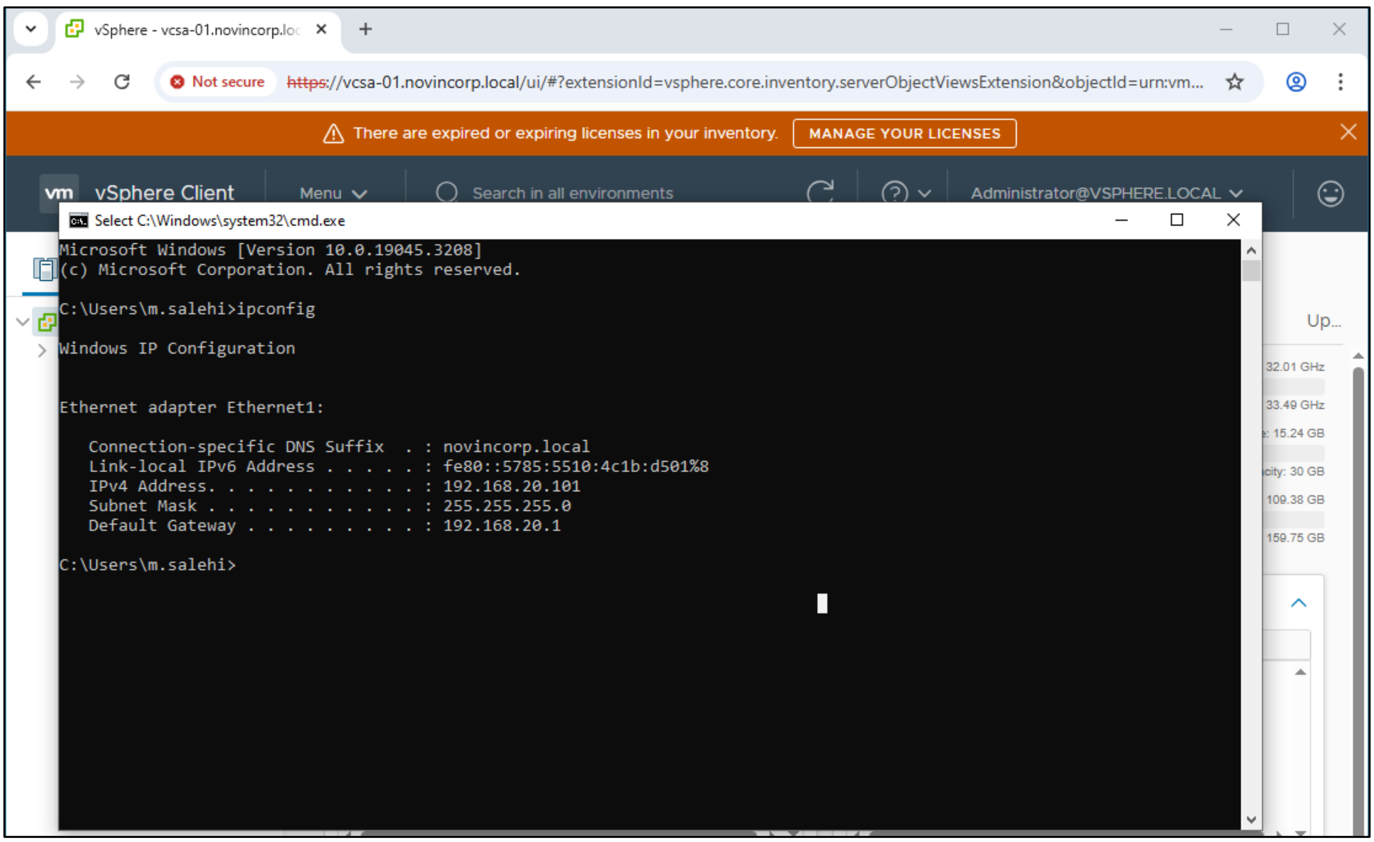




Sales and Staff can’t see each other



Only It admin have access to http management web console of hosts



Other users don't have access to web management consols.

```
C:\Windows\system32\cmd.exe

C:\Users\r.hosseini>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : novincorp.local
   Link-local IPv6 Address . . . . . : fe80::86ed:7191:c111:7018%11
   IPv4 Address. . . . . . . . . . . : 192.168.10.100
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1

C:\Users\r.hosseini>ping vcsa-01.novincorp.local

Pinging vcsa-01.novincorp.local [192.168.60.15] with 32 bytes of data:
Reply from 192.168.10.1: Destination net unreachable.
Reply from 192.168.10.1: Destination net unreachable.
Reply from 192.168.10.1: Destination net unreachable.
Reply from 192.168.10.1: Destination net unreachable.

Ping statistics for 192.168.60.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\r.hosseini>
```

17. Troubleshooting and Engineering Lessons

17.1 GNS3 / IOSv instability

```
Example symptoms observed:
%SYS-3-CPUHOG
CPU exception occurred: Overflow(4)
Unexpected exception to CPU
Writing crashinfo to flash0:crashinfo_*.
Detected TX Unit Hang on a nested TrueNAS E1000 path
```

The lab initially associated performance and storage failures with iSCSI/TrueNAS. Packet capture and direct-VMnet testing showed that the iSCSI datastore could be created reliably when the problematic virtual switching path was bypassed. The Cisco emulation layer was moved to EVE-NG 6.2 and a different IOL image was selected. This improved throughput behavior and removed the repeatable IOSv crash pattern.

17.2 ACL return-path lesson

The initial inter-VLAN ACL policy allowed IT to initiate connections to Sales/Staff but blocked the reverse packet because ACLs are stateless. The fix was to explicitly allow the return direction before the deny entry.

17.3 Windows Host routing lesson

The physical Workstation had multiple VMware adapters and multiple default routes, including a VPN-style /1 routing pattern. A specific route for 192.168.40.0/24 via 192.168.60.1 restored Host-to-DC reachability without disturbing Internet access.

This route was written to access web-ui administrator console of host dns records via the main host.

17.4 iSCSI write path lesson

iSCSI discovery and LUN visibility are not sufficient validation. The path must be proven writable by creating the VMFS datastore and running a workload on it. Direct-VMnet testing isolated the earlier failure to the emulated switching path.

The TrueNAS VM was initially using an E1000 virtual NIC and exhibited TX Unit Hang behavior on the storage path. Replacing the virtual NIC with VMXNET3 eliminated the observed storage-path instability in the lab.

18. Performance Validation

Same VLAN iperf3:

Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\r.hosseini> cd C:\iperf3
PS C:\iperf3> .\iperf3.exe -c 192.168.10.101
Connecting to host 192.168.10.101, port 5201
[  5] local 192.168.10.100 port 50274 connected to 192.168.10.101 port 5201
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.00   sec   634 MBytes  5.32 Gbits/sec
[  5]   1.00-2.00   sec   774 MBytes  6.50 Gbits/sec
[  5]   2.00-3.00   sec   884 MBytes  7.42 Gbits/sec
[  5]   3.00-4.00   sec   806 MBytes  6.76 Gbits/sec
[  5]   4.00-5.00   sec   831 MBytes  6.97 Gbits/sec
[  5]   5.00-6.00   sec   821 MBytes  6.88 Gbits/sec
[  5]   6.00-7.00   sec   785 MBytes  6.58 Gbits/sec
[  5]   7.00-8.00   sec   783 MBytes  6.57 Gbits/sec
[  5]   8.00-9.00   sec   768 MBytes  6.44 Gbits/sec
[  5]   9.00-10.00  sec   788 MBytes  6.61 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.00  sec  7.69 GBytes  6.61 Gbits/sec                  sender
[  5]   0.00-10.10  sec  7.69 GBytes  6.54 Gbits/sec                  receiver

iperf Done.
PS C:\iperf3>
```

ff

Cross VLAN:

```
PS C:\iperf3> .\iperf3.exe -c 192.168.20.101
Connecting to host 192.168.20.101, port 5201
[  5] local 192.168.10.100 port 50276 connected to 192.168.20.101 port 5201
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.02   sec  3.50 MBytes  28.9 Mbits/sec
[  5]   1.02-2.01   sec  8.00 MBytes  67.7 Mbits/sec
[  5]   2.01-3.02   sec  3.25 MBytes  26.8 Mbits/sec
[  5]   3.02-4.01   sec  5.12 MBytes  43.5 Mbits/sec
[  5]   4.01-5.00   sec  5.00 MBytes  42.4 Mbits/sec
[  5]   5.00-6.00   sec  8.00 MBytes  67.0 Mbits/sec
[  5]   6.00-7.00   sec  8.25 MBytes  69.4 Mbits/sec
[  5]   7.00-8.00   sec  10.6 MBytes  89.1 Mbits/sec
[  5]   8.00-9.01   sec  6.75 MBytes  56.4 Mbits/sec
[  5]   9.01-10.01  sec  7.50 MBytes  62.9 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.01  sec  66.0 MBytes  55.3 Mbits/sec                  sender
[  5]   0.00-10.03  sec  65.8 MBytes  55.0 Mbits/sec                  receiver

iperf Done.
PS C:\iperf3>
```

The discrepancy triggered packet-level troubleshooting and ultimately led to replacing the unstable IOSv emulation path.

19. Resume-Ready Project Summary

Project title

Design and Implementation of a Virtualized Small-Enterprise Data Center and Segmented Cisco Network

Suggested resume bullets

- Designed and implemented a nested VMware lab with two ESXi 6.7 hosts, VCSA 6.7, TrueNAS iSCSI shared storage and Windows Server 2016 infrastructure services.
- Designed a segmented Cisco L3/L2 network using seven VLANs for users, servers, storage, management and vMotion; implemented centralized inter-VLAN routing and DHCP relay.
- Implemented role-based Active Directory security using Global-to-Domain-Local group nesting, NTFS/SMB permissions, department shares and GPO-based user configuration.
- Built a shared iSCSI VMFS datastore on TrueNAS and validated vCenter-managed workloads and ESXi HA recovery behavior.
- Implemented ACL-based segmentation so Sales/Staff can reach the DC while IT administration is selectively permitted to infrastructure management VLANs.
- Troubleshot nested virtual networking using Wireshark, ICMP, TCP/445 and iperf3; migrated the Cisco emulation layer from GNS3 to EVE-NG after reproducible IOSv instability.
- Prepared and partially implemented CCNA-level hardening controls including SSH-only management, VTY source restrictions.
- PortFast/BPDU Guard, with DHCP Snooping/DAI planned as the next security-hardening step.
- This project was built exclusively as a personal educational lab and is not intended or validated for production use.

Portfolio narrative

The strongest portfolio angle is not only that the lab works; it demonstrates a complete engineering workflow: requirements analysis, topology design, segmentation, implementation, validation, fault isolation, evidence collection and iterative redesign when a virtualization/emulation component proved unstable.

20. Next Phase: Internet Edge Firewall and Remote VPN

The planned next phase is to add a single virtual firewall at the network edge rather than introducing a second router solely for NAT. The Cisco core remains responsible for internal inter-VLAN routing; the firewall provides Internet NAT, stateful filtering, logging and remote-access VPN.

```
Internet
   |
[Firewall VM]
   |  Transit
[Core L3]
   |-- VLAN 10 Sales
   |-- VLAN 20 IT
   |-- VLAN 30 Staff
   |-- VLAN 40 Servers
   |-- VLAN 50 Storage
   |-- VLAN 60 Management
   `-- VLAN 70 vMotion

Remote VPN users
   -> Firewall VPN pool
   -> controlled access to internal resources
```

A single firewall can provide Internet gateway, NAT and VPN in this lab. A second router/firewall should only be added later if the portfolio objective explicitly includes edge-router/firewall separation or HA firewall design.

Appendix A - Command Capture Index

Area	Command	Capture from
VLAN / ports	show vlan brief	Core + Access
Trunks	show interfaces trunk	Core + Access
SVIs	show ip interface brief	Core
Routing	show ip route	Core
ACLs	show access-lists	Core
ACL attachment	show run | section interface Vlan	Core
STP	show spanning-tree vlan 10 / 20 / 30 / 40	Core + Access
Interface errors	show interfaces <port>	Core + Access
SSH	show ip ssh	Core + Access
Users / VTY	show users / show line	Core + Access
ESXi NICs	esxcli network nic list	ESXi01/02
ESXi VMkernel	esxcli network ip interface list	ESXi01/02
iSCSI adapters	esxcli iscsi adapter list	ESXi01/02
Storage paths	esxcli storage core path list	ESXi01/02
Datastores	esxcli storage filesystem list	ESXi01/02
AD result	gpresult /r	DC + sample clients
DNS	nslookup <hostname> <DC-IP>	Admin workstation
Network throughput	iperf3.exe -s / iperf3.exe -c ... -t 30 -i 1	DC + clients